

This has reduced our problem to computing the derivatives of x^5 , which is easy, and of $(3 + x^6)^{1/2}$ and $(4 + x^2)^{-3}$, both of which can be done by the chain rule. Doing so,

$$\begin{aligned} \frac{d}{dx} [x^5 (3 + x^6)^{1/2} (4 + x^2)^{-3}] &= \overbrace{\frac{d}{dx} [x^5]}^{5x^4} \cdot (3 + x^6)^{1/2} \cdot (4 + x^2)^{-3} \\ &\quad + x^5 \cdot \overbrace{\frac{d}{dx} [(3 + x^6)^{1/2}]}^{\frac{1}{2}(3+x^6)^{-1/2} \cdot 6x^5} \cdot (4 + x^2)^{-3} \\ &\quad + x^5 \cdot (3 + x^6)^{1/2} \cdot \overbrace{\frac{d}{dx} [(4 + x^2)^{-3}]}^{-3(4+x^2)^{-4} \cdot 2x} \end{aligned}$$

- Now we can clean things up in a sneaky way by observing
 - differentiating x^5 , to get $5x^4$, is the same as multiplying x^5 by $\frac{5}{x}$, and
 - differentiating $(3 + x^6)^{1/2}$ to get $\frac{1}{2}(3 + x^6)^{-1/2} \cdot 6x^5$ is the same as multiplying $(3 + x^6)^{1/2}$ by $\frac{3x^5}{3+x^6}$, and
 - differentiating $(4 + x^2)^{-3}$ to get $-3(4 + x^2)^{-4} \cdot 2x$ is the same as multiplying $(4 + x^2)^{-3}$ by $-\frac{6x}{4+x^2}$.

Using these sneaky tricks we can write our solution quite neatly:

$$\frac{d}{dx} \cos \left(\frac{x^5 \sqrt{3 + x^6}}{(4 + x^2)^3} \right) = -\sin \left(\frac{x^5 \sqrt{3 + x^6}}{(4 + x^2)^3} \right) \frac{x^5 \sqrt{3 + x^6}}{(4 + x^2)^3} \left\{ \frac{5}{x} + \frac{3x^5}{3 + x^6} - \frac{6x}{4 + x^2} \right\}$$

- This method of cleaning up the derivative of a messy product is actually something more systematic in disguise — namely logarithmic differentiation. We will come to this later.

Example 2.9.16

2.10 ▲ The Natural Logarithm

The chain rule opens the way to understanding derivatives of more complicated function. Not only compositions of known functions as we have seen the examples of the previous section, but also functions which are defined implicitly.

Consider the logarithm base e — $\log_e(x)$ is the power that e must be raised to to give x . That is, $\log_e(x)$ is defined by

$$e^{\log_e x} = x$$

i.e. — it is the inverse of the exponential function with base e . Since this choice of base works so cleanly and easily with respect to differentiation, this base turns out to be (arguably) the most natural choice for the base of the logarithm. And as we saw in our whirlwind review of logarithms in Section 2.7, it is easy to use logarithms of one base to compute logarithms with another base:

$$\log_q x = \frac{\log_e x}{\log_e q}$$

So we are (relatively) free to choose a base which is convenient for our purposes.

The logarithm with base e , is called the “natural logarithm”. The “naturalness” of logarithms base e is exactly that this choice of base works very nicely in calculus (and so wider mathematics) in ways that other bases do not³⁴. There are several different “standard” notations for the logarithm base e ;

$$\log_e x = \log x = \ln x.$$

We recommend that you be able to recognise all of these.

In this text we will write the natural logarithm as “log” with no base. The reason for this choice is that base e is the standard choice of base for logarithms in mathematics³⁵. The natural logarithm inherits many properties of general logarithms³⁶. So, for all $x, y > 0$ the following hold:

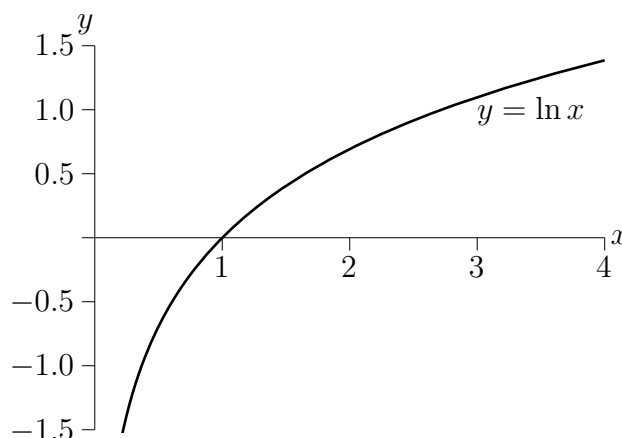
- $e^{\log x} = x$,
- for any real number X , $\log(e^X) = X$,
- for any $a > 1$, $\log_a x = \frac{\log x}{\log a}$ and $\log x = \frac{\log_a x}{\log_a e}$
- $\log 1 = 0$, $\log e = 1$
- $\log(xy) = \log x + \log y$
- $\log\left(\frac{x}{y}\right) = \log x - \log y$, $\log\left(\frac{1}{y}\right) = -\log y$
- $\log(x^X) = X \log x$
- $\lim_{x \rightarrow \infty} \log x = \infty$, $\lim_{x \rightarrow 0} \log x = -\infty$

And finally we should remember that $\log x$ has domain (i.e. is defined for) $x > 0$ and range (i.e. takes all of the values in) the set of all real numbers.

34 The interested reader should head to Wikipedia and look up the natural logarithm.

35 In other disciplines other bases are natural; in computer science, since numbers are stored in binary it makes sense to use the binary logarithm — i.e. base 2. While in some sciences and finance, it makes sense to use the decimal logarithm — i.e. base 10.

36 Again take a quick look at the whirlwind review of logarithms in Section 2.7.

Figure 2.10.1.

To compute the derivative of $\log x$ we could attempt to start with the limit definition of the derivative

$$\begin{aligned}\frac{d}{dx} \log x &= \lim_{h \rightarrow 0} \frac{\log(x+h) - \log(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\log((x+h)/x)}{h} \\ &= \text{um...}\end{aligned}$$

This doesn't look good. But all is not lost — we have the chain rule, and we know that the logarithm satisfies the equation:

$$x = e^{\log x}$$

Since both sides of the equation are the same function, both sides of the equation have the same derivative. i.e. we are using³⁷

$$\text{if } f(x) = g(x) \text{ for all } x, \text{ then } f'(x) = g'(x)$$

So now differentiate both sides:

$$\frac{d}{dx} x = \frac{d}{dx} e^{\log x}$$

The left-hand side is easy, and the right-hand side we can process using the chain rule with $f(u) = e^u$ and $u = \log x$.

$$\begin{aligned}1 &= \frac{df}{du} \cdot \frac{du}{dx} \\ &= e^u \cdot \underbrace{\frac{d}{dx} \log x}_{\text{what we want to compute}}\end{aligned}$$

³⁷ Notice that just because the derivatives are the same, doesn't mean the original functions are the same. Both $f(x) = x^2$ and $g(x) = x^2 + 3$ have derivative $f'(x) = g'(x) = 2x$, but $f(x) \neq g(x)$.

Recall that $e^u = e^{\log x} = x$, so

$$1 = x \cdot \underbrace{\frac{d}{dx} \log x}_{\text{now what?}}$$

We can now just rearrange this equation to make the thing we want the subject:

$$\frac{d}{dx} \log x = \frac{1}{x}$$

Thus we have proved:

Theorem 2.10.1.

$$\frac{d}{dx} \log x = \frac{1}{x}$$

where $\log x$ is the logarithm base e .

Example 2.10.2

Let $f(x) = \log 3x$. Find $f'(x)$.

There are two ways to approach this — we can simplify then differentiate, or differentiate and then simplify. Neither is difficult.

- Simplify and then differentiate:

$$\begin{aligned} f(x) &= \log 3x && \text{log of a product} \\ &= \log 3 + \log x \\ f'(x) &= \frac{d}{dx} \log 3 + \frac{d}{dx} \log x \\ &= \frac{1}{x}. \end{aligned}$$

- Differentiation and then simplify:

$$\begin{aligned} f'(x) &= \frac{d}{dx} \log(3x) && \text{chain rule} \\ &= \frac{1}{3x} \cdot 3 \\ &= \frac{1}{x} \end{aligned}$$

Example 2.10.2

Example 2.10.3 (The derivative of $\log cx$)

Notice that we can extend the previous example for any positive constant — not just 3. Let $c > 0$ be a constant, then

$$\begin{aligned}\frac{d}{dx} \log cx &= \frac{d}{dx} (\log c + \log x) \\ &= \frac{1}{x}\end{aligned}$$

Example 2.10.3

Example 2.10.4 (The derivative of $\log |x|$)

We can push this further still. Let $g(x) = \log |x|$, then³⁸

- If $x > 0$, $|x| = x$ and so

$$g'(x) = \frac{d}{dx} \log x = \frac{1}{x}$$

- If $x < 0$ then $|x| = -x$. If $|h|$ is strictly smaller than $|x|$, then we also have that $x + h < 0$ and $|x + h| = -(x + h) = |x| - h$. Write $X = |x|$ and $H = -h$. Then, by the definition of the derivative,

$$\begin{aligned}g'(x) &= \lim_{h \rightarrow 0} \frac{\log |x + h| - \log |x|}{h} = \lim_{h \rightarrow 0} \frac{\log(|x| - h) - \log |x|}{h} \\ &= \lim_{H \rightarrow 0} \frac{\log(X + H) - \log X}{-H} = - \lim_{H \rightarrow 0} \frac{\log(X + H) - \log X}{H} \\ &= - \frac{d}{dX} \log X = - \frac{1}{X} = - \frac{1}{|x|} \\ &= \frac{1}{x}\end{aligned}$$

- Since $\log 0$ is undefined, $g'(0)$ does not exist.

Putting this together gives:

$$\frac{d}{dx} \log |x| = \frac{1}{x}$$

Example 2.10.4

Example 2.10.5 (The derivative of x^a)

Just after Corollary [2.6.17](#), we said that we would, in the future, find the derivative of x^a

³⁸ It's probably a good moment to go back and look at Example [2.2.10](#).

for all real numbers. The future is here. Let $x > 0$ and a be any real number. Exponentiating both sides of $\log(x^a) = a \log x$ gives us $x^a = e^{a \log x}$ and then

$$\begin{aligned} \frac{d}{dx} x^a &= \frac{d}{dx} e^{a \log x} = e^{a \log x} \frac{d}{dx} (a \log x) && \text{by the chain rule} \\ &= \frac{a}{x} e^{a \log x} = \frac{a}{x} x^a \\ &= a x^{a-1} \end{aligned}$$

as expected.

Example 2.10.5

We can extend Theorem 2.10.1 to compute the derivative of logarithms of other bases in a straightforward way. Since for any positive $a \neq 1$:

$$\begin{aligned} \log_a x &= \frac{\log x}{\log a} = \frac{1}{\log a} \cdot \log x && \text{since } a \text{ is a constant} \\ \frac{d}{dx} \log_a x &= \frac{1}{\log a} \cdot \frac{1}{x} \end{aligned}$$

►► Back to $\frac{d}{dx} a^x$

We can also now finally get around to computing the derivative of a^x (which we started to do back in Section 2.7).

$$\begin{aligned} f(x) &= a^x && \text{take log of both sides} \\ \log f(x) &= x \log a && \text{exponentiate both sides base } e \\ f(x) &= e^{x \log a} && \text{chain rule} \\ f'(x) &= e^{x \log a} \cdot \log a \\ &= a^x \cdot \log a \end{aligned}$$

Notice that we could have also done the following:

$$\begin{aligned} f(x) &= a^x && \text{take log of both sides} \\ \log f(x) &= x \log a && \text{differentiate both sides} \\ \frac{d}{dx} (\log f(x)) &= \log a \end{aligned}$$

We then process the left-hand side using the chain rule

$$\begin{aligned} f'(x) \cdot \frac{1}{f(x)} &= \log a \\ f'(x) &= f(x) \cdot \log a = a^x \cdot \log a \end{aligned}$$

We will see $\frac{d}{dx} \log f(x)$ more below in the subsection on “logarithmic differentiation”.

To summarise the results above:

Corollary 2.10.6.

$$\begin{aligned}\frac{d}{dx} a^x &= \log a \cdot a^x && \text{for any } a > 0 \\ \frac{d}{dx} \log_a x &= \frac{1}{x \cdot \log a} && \text{for any } a > 0, a \neq 1\end{aligned}$$

where $\log x$ is the natural logarithm.

Recall that we need the caveat $a \neq 1$ because the logarithm base 1 is not well defined. This is because $1^x = 1$ for any x . We do not need a similar caveat for the derivative of the exponential because we know (recall Example 2.7.1)

$$\begin{aligned}\frac{d}{dx} 1^x &= \frac{d}{dx} 1 = 0 && \text{while the above corollary tells us} \\ &= \log 1 \cdot 1^x = 0 \cdot 1 = 0.\end{aligned}$$

► Logarithmic Differentiation

We want to go back to some previous slightly messy examples (Examples 2.6.6 and 2.6.18) and now show you how they can be done more easily.

Example 2.10.7

Consider again the derivative of the product of 3 functions:

$$P(x) = F(x) \cdot G(x) \cdot H(x)$$

Start by taking the logarithm of both sides:

$$\begin{aligned}\log P(x) &= \log (F(x) \cdot G(x) \cdot H(x)) \\ &= \log F(x) + \log G(x) + \log H(x)\end{aligned}$$

Notice that the product of functions on the right-hand side has become a sum of functions. Differentiating sums is much easier than differentiating products. So when we differentiate we have

$$\frac{d}{dx} \log P(x) = \frac{d}{dx} \log F(x) + \frac{d}{dx} \log G(x) + \frac{d}{dx} \log H(x)$$

A quick application of the chain rule shows that $\frac{d}{dx} \log f(x) = f'(x)/f(x)$:

$$\frac{P'(x)}{P(x)} = \frac{F'(x)}{F(x)} + \frac{G'(x)}{G(x)} + \frac{H'(x)}{H(x)}$$

Multiply through by $P(x) = F(x)G(x)H(x)$:

$$\begin{aligned}P'(x) &= \left(\frac{F'(x)}{F(x)} + \frac{G'(x)}{G(x)} + \frac{H'(x)}{H(x)} \right) \cdot F(x)G(x)H(x) \\ &= F'(x)G(x)H(x) + F(x)G'(x)H(x) + F(x)G(x)H'(x)\end{aligned}$$

which is what found in Example 2.6.6 by repeated application of the product rule. The above generalises quite easily to more than 3 functions.

Example 2.10.7

This same trick of “take a logarithm and then differentiate” — or logarithmic differentiation — will work any time you have a product (or ratio) of functions.

Example 2.10.8

Let’s use logarithmic differentiation on the function from Example 2.6.18:

$$f(x) = \frac{(\sqrt{x} - 1)(2 - x)(1 - x^2)}{\sqrt{x}(3 + 2x)}$$

Beware however, that we may only take the logarithm of positive numbers, and this $f(x)$ is often negative. For example, if $1 < x < 2$, the factor $(1 - x^2)$ in the definition of $f(x)$ is negative while all of the other factors are positive, so that $f(x) < 0$. None-the-less, we can use logarithmic differentiation to find $f'(x)$, by exploiting the observation that $\frac{d}{dx} \log |f(x)| = \frac{f'(x)}{f(x)}$. (To see this, use the chain rule and Example 2.10.4.) So we take the logarithm of $|f(x)|$ and expand.

$$\begin{aligned} \log |f(x)| &= \log \frac{|\sqrt{x} - 1| |2 - x| |1 - x^2|}{\sqrt{x} |3 + 2x|} \\ &= \log |\sqrt{x} - 1| + \log |2 - x| + \log |1 - x^2| - \underbrace{\log(\sqrt{x})}_{=\frac{1}{2} \log x} - \log |3 + 2x| \end{aligned}$$

Now we can essentially just differentiate term-by-term:

$$\begin{aligned} \frac{d}{dx} \log |f(x)| &= \frac{d}{dx} \left(\log |\sqrt{x} - 1| + \log |2 - x| + \log |1 - x^2| - \frac{1}{2} \log(x) - \log |3 + 2x| \right) \\ \frac{f'(x)}{f(x)} &= \frac{1/(2\sqrt{x})}{\sqrt{x} - 1} + \frac{-1}{2 - x} + \frac{-2x}{1 - x^2} - \frac{1}{2x} - \frac{2}{3 + 2x} \\ f'(x) &= f(x) \cdot \left(\frac{1}{2\sqrt{x}(\sqrt{x} - 1)} - \frac{1}{2 - x} - \frac{2x}{1 - x^2} - \frac{1}{2x} - \frac{2}{3 + 2x} \right) \\ &= \frac{(\sqrt{x} - 1)(2 - x)(1 - x^2)}{\sqrt{x}(3 + 2x)} \cdot \left(\frac{1}{2\sqrt{x}(\sqrt{x} - 1)} - \frac{1}{2 - x} - \frac{2x}{1 - x^2} - \frac{1}{2x} - \frac{2}{3 + 2x} \right) \end{aligned}$$

just as we found previously.

Example 2.10.8

2.11 ▲ Implicit Differentiation

Implicit differentiation is a simple trick that is used to compute derivatives of functions either

- when you don't know an explicit formula for the function, but you know an equation that the function obeys or
- even when you have an explicit, but complicated, formula for the function, and the function obeys a simple equation.

The trick is just to differentiate both sides of the equation and then solve for the derivative we are seeking. In fact we have already done this, without using the name “implicit differentiation”, when we found the derivative of $\log x$ in the previous section. There we knew that the function $f(x) = \log x$ satisfied the equation $e^{f(x)} = x$ for all x . That is, the functions $e^{f(x)}$ and x are in fact the same function and so have the same derivative. So we had

$$\frac{d}{dx}e^{f(x)} = \frac{d}{dx}x = 1$$

We then used the chain rule to get $\frac{d}{dx}e^{f(x)} = e^{f(x)}f'(x)$, which told us that $f'(x)$ obeys the equation

$$e^{f(x)}f'(x) = 1 \quad \text{and we can now solve for } f'(x)$$

$$f'(x) = e^{-f(x)} = e^{-\log x} = \frac{1}{x}.$$

The typical way to get used to implicit differentiation is to play with problems involving tangent lines to curves. So here are a few examples finding the equations of tangent lines to curves. Recall, from Theorem 2.3.2, that, in general, the tangent line to the curve $y = f(x)$ at (x_0, y_0) is $y = f(x_0) + f'(x_0)(x - x_0) = y_0 + f'(x_0)(x - x_0)$.

Example 2.11.1

Find the equation of the tangent line to $y = y^3 + xy + x^3$ at $x = 1$.

This is a very standard sounding example, but made a little complicated by the fact that the curve is given by a cubic equation — which means we cannot solve directly for y in terms of x or vice versa. So we really do need implicit differentiation.

- First notice that when $x = 1$ the equation, $y = y^3 + xy + x^3$, of the curve simplifies to $y = y^3 + y + 1$ or $y^3 = -1$, which we can solve³⁹: $y = -1$. So we know that the curve passes through $(1, -1)$ when $x = 1$.
- Now, to find the slope of the tangent line at $(1, -1)$, pretend that our curve is $y = f(x)$ so that $f(x)$ obeys

$$f(x) = f(x)^3 + xf(x) + x^3$$

for all x . Differentiating both sides gives

$$f'(x) = 3f(x)^2f'(x) + f(x) + xf'(x) + 3x^2$$

39 This type of luck rarely happens in the “real world”. But it happens remarkably frequently in textbooks, problem sets and tests.

- At this point we could isolate for $f'(x)$ and write it in terms of $f(x)$ and x , but since we only want answers when $x = 1$, let us substitute in $x = 1$ and $f(1) = -1$ (since the curve passes through $(1, -1)$) and clean things up before doing anything else.
- Subbing in $x = 1$, $f(1) = -1$ gives

$$f'(1) = 3f'(1) - 1 + f'(1) + 3 \quad \text{and so } f'(1) = -\frac{2}{3}$$

- The equation of the tangent line is

$$y = y_0 + f'(x_0)(x - x_0) = -1 - \frac{2}{3}(x - 1) = -\frac{2}{3}x - \frac{1}{3}$$

We can further clean up the equation of the line to write it as $2x + 3y = -1$.

Example 2.11.1

In the previous example we replace y by $f(x)$ in the middle of the computation. We don't actually have to do this. When we are writing out our solution we can remember that y is a function of x . So we can start with

$$y = y^3 + xy + x^3$$

and differentiate remembering that $y \equiv y(x)$

$$y' = 3y^2y' + xy' + y + 3x^2$$

And now substitute $x = 1, y = -1$ to get

$$\begin{aligned} y'(1) &= 3 \cdot y'(1) + y'(1) - 1 + 3 && \text{and so} \\ y'(1) &= -\frac{2}{3} \end{aligned}$$

The next one is at the same time a bit easier (because it is a quadratic) and a bit harder (because we are asked for the tangent at a general point on the curve, not a specific one).

Example 2.11.2

Let (x_0, y_0) be a point on the ellipse $3x^2 + 5y^2 = 7$. Find the equation for the tangent lines when $x = 1$ and y is positive. Then find an equation for the tangent line to the ellipse at a general point (x_0, y_0) .

Since we are not given a specific point x_0 we are going to have to be careful with the second half of this question.

- When $x = 1$ the equation simplifies to

$$\begin{aligned} 3 + 5y^2 &= 7 \\ 5y^2 &= 4 \\ y &= \pm \frac{2}{\sqrt{5}}. \end{aligned}$$

We are only interested in positive y , so our point on the curve is $(1, 2/\sqrt{5})$.

- Now we use implicit differentiation to find $\frac{dy}{dx}$ at this point. First we pretend that we have solved the curve explicitly, for some interval of x 's, as $y = f(x)$. The equation becomes

$$\begin{aligned} 3x^2 + 5f(x)^2 &= 7 && \text{now differentiate} \\ 6x + 10f(x)f'(x) &= 0 \\ f'(x) &= -\frac{3x}{5f(x)} \end{aligned}$$

- When $x = 1, y = 2/\sqrt{5}$ this becomes

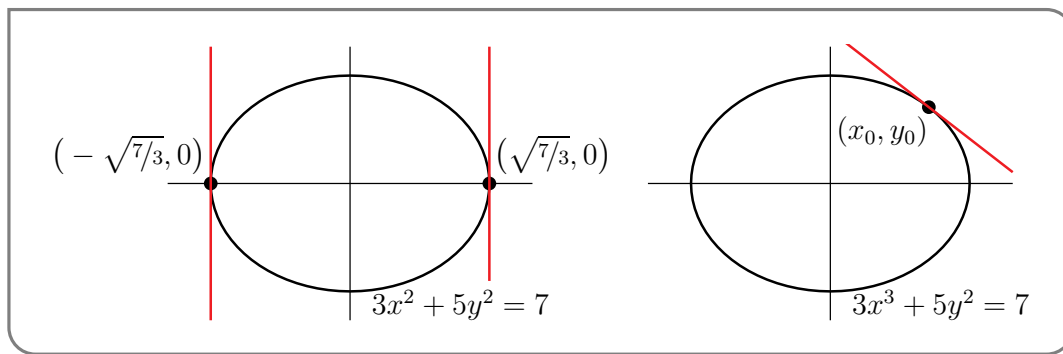
$$f'(1) = -\frac{3}{5 \cdot 2/\sqrt{5}} = -\frac{3}{2\sqrt{5}}$$

So the tangent line passes through $(1, 2/\sqrt{5})$ and has slope $-\frac{3}{2\sqrt{5}}$. Hence the tangent line has equation

$$\begin{aligned} y &= y_0 + f'(x_0)(x - x_0) \\ &= \frac{2}{\sqrt{5}} - \frac{3}{2\sqrt{5}}(x - 1) \\ &= \frac{7 - 3x}{2\sqrt{5}} && \text{or equivalently} \\ 3x + 2\sqrt{5}y &= 7 \end{aligned}$$

Now we should go back and do the same but for a general point on the curve (x_0, y_0) :

- A good first step here is to sketch the curve. Since this is an ellipse, it is pretty straight-forward.



- Notice that there are two points on the ellipse — the extreme right and left points $(x_0, y_0) = \pm(\sqrt{7/3}, 0)$ — at which the tangent line is vertical. In those two cases, the tangent line is just $x = x_0$.
- Since this is a quadratic for y , we could solve it explicitly to get

$$y = \pm\sqrt{\frac{7 - 3x^2}{5}}$$

and choose the positive or negative branch as appropriate. Then we could differentiate to find the slope and put things together to get the tangent line.

But even in this relatively easy case, it is computationally cleaner, and hence less vulnerable to mechanical errors, to use implicit differentiation. So that's what we'll do.

- Now we could again “pretend” that we have solved the equation for the ellipse for $y = f(x)$ near (x_0, y_0) , but let's not do that. Instead (as we did just before this example) just remember that when we differentiate y is really a function of x . So starting from

$$\begin{aligned} 3x^2 + 5y^2 &= 7 && \text{differentiating gives} \\ 6x + 5 \cdot 2y \cdot y' &= 0 \end{aligned}$$

We can then solve this for y' :

$$y' = -\frac{3x}{5y}$$

where y' and y are both functions of x .

- Hence at the point (x_0, y_0) we have

$$y'|_{(x_0, y_0)} = -\frac{3x_0}{5y_0}$$

This is the slope of the tangent line at (x_0, y_0) and so its equation is

$$\begin{aligned} y &= y_0 + y' \cdot (x - x_0) \\ &= y_0 - \frac{3x_0}{5y_0}(x - x_0) \end{aligned}$$

We can simplify this by multiplying through by $5y_0$ to get

$$5y_0y = 5y_0^2 - 3x_0x + 3x_0^2$$

We can clean this up more by moving all the terms that contain x or y to the left-hand side and everything else to the right:

$$3x_0x + 5y_0y = 3x_0^2 + 5y_0^2$$

But there is one more thing we can do, our original equation is $3x^2 + 5y^2 = 7$ for all points on the curve, so we know that $3x_0^2 + 5y_0^2 = 7$. This cleans up the right-hand side.

$$3x_0x + 5y_0y = 7$$

- In deriving this formula for the tangent line at (x_0, y_0) we have assumed that $y_0 \neq 0$. But in fact the final answer happens to also work when $y_0 = 0$ (which means $x_0 = \pm\sqrt{7/3}$), so that the tangent line is $x = x_0$.

We can also check that our answer for general (x_0, y_0) reduces to our answer for $x_0 = 1$.

- When $x_0 = 1$ we worked out that $y_0 = 2/\sqrt{5}$.
- Plugging this into our answer above gives

$$\begin{array}{ll} 3x_0x + 5y_0y = 7 & \text{sub in } (x_0, y_0) = (1, 2/\sqrt{5}) : \\ 3x + 5\frac{2}{\sqrt{5}}y = 7 & \text{clean up a little} \\ 3x + 2\sqrt{5}y = 7 & \end{array}$$

as required.

Example 2.11.2

Example 2.11.3

At which points does the curve $x^2 - xy + y^2 = 3$ cross the x -axis? Are the tangent lines to the curve at those points parallel?

This is a 2 part question — first the x -intercepts and then we need to examine tangent lines.

- Finding where the curve crosses the x -axis is straight forward. It does so when $y = 0$. This means x satisfies

$$x^2 - x \cdot 0 + 0^2 = 3 \quad \text{so } x = \pm\sqrt{3}.$$

So the curve crosses the x -axis at two points $(\pm\sqrt{3}, 0)$.

- Now we need to find the tangent lines at those points. But we don't actually need the lines, just their slopes. Again we can pretend that near one of those points the curve is $y = f(x)$. Applying $\frac{d}{dx}$ to both sides of $x^2 - xf(x) + f(x)^2 = 3$ gives

$$2x - f(x) - xf'(x) + 2f(x)f'(x) = 0$$

etc etc.

- But let us stop “pretending”. Just make sure we remember that y is a function of x when we differentiate:

$$\begin{array}{ll} x^2 - xy + y^2 = 3 & \text{start with the curve, and differentiate} \\ 2x - xy' - y + 2yy' = 0 & \end{array}$$

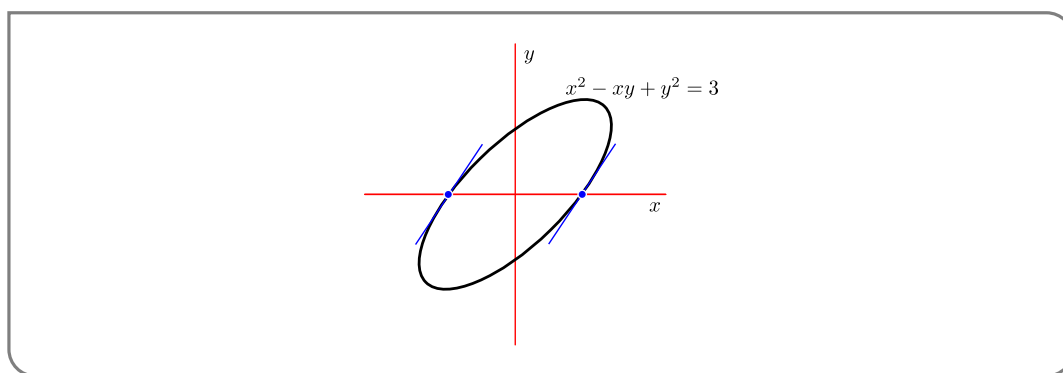
Now substitute in the first point, $x = +\sqrt{3}, y = 0$:

$$\begin{array}{l} 2\sqrt{3} - \sqrt{3}y' + 0 = 0 \\ y' = 2 \end{array}$$

And now do the second point $x = -\sqrt{3}, y = 0$:

$$\begin{array}{l} -2\sqrt{3} + \sqrt{3}y' + 0 = 0 \\ y' = 2 \end{array}$$

Thus the slope is the same at $x = \sqrt{3}$ and $x = -\sqrt{3}$ and the tangent lines are parallel.



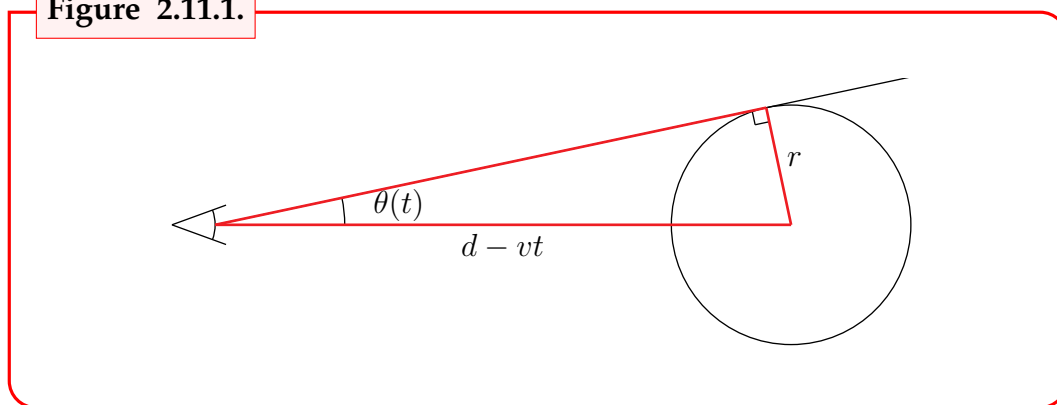
Example 2.11.3

Okay — let's get away from curves and do something a little different.

Example 2.11.4

You are standing at the origin. At time zero a pitcher throws a ball at your head⁴⁰.

Figure 2.11.1.



The position of the (centre of the) ball at time t is $x(t) = d - vt$, where d is the distance from your head to the pitcher's mound and v is the ball's velocity. Your eye sees the ball filling⁴¹ an angle $2\theta(t)$ with

$$\sin(\theta(t)) = \frac{r}{d - vt}$$

where r is the radius of the baseball. The question is “How fast is θ growing at time t ?” That is, what is $\frac{d\theta}{dt}$?

- We don't know (yet) how to solve this equation to find $\theta(t)$ explicitly. So we use implicit differentiation.
- To do so we apply $\frac{d}{dt}$ to both sides of our equation. This gives

$$\cos(\theta(t)) \cdot \theta'(t) = \frac{rv}{(d - vt)^2}$$

40 It seems that it is not a friendly game today.

41 This is the “visual angle” or “angular size”.

- Then we solve for $\theta'(t)$:

$$\theta'(t) = \frac{rv}{(d - vt)^2 \cos(\theta(t))}$$

- As is often the case, when using implicit differentiation, this answer is not very satisfying because it contains $\theta(t)$, for which we still do not have an explicit formula. However in this case we can get an explicit formula for $\cos(\theta(t))$, without having an explicit formula for $\theta(t)$, just by looking at the right-angled triangle in Figure 2.11.1 above.
- The hypotenuse of that triangle has length $d - vt$. By Pythagoras, the length of the side of the triangle adjacent of the angle $\theta(t)$ is $\sqrt{(d - vt)^2 - r^2}$. So

$$\cos(\theta(t)) = \frac{\sqrt{(d - vt)^2 - r^2}}{d - vt}$$

and

$$\theta'(t) = \frac{rv}{(d - vt)\sqrt{(d - vt)^2 - r^2}}$$

Example 2.11.4

Okay — just one more tangent-to-the-curve example and then we'll go on to something different.

Example 2.11.5

Let (x_0, y_0) be a point on the astroid⁴²

$$x^{2/3} + y^{2/3} = 1.$$

Find an equation for the tangent line to the astroid at (x_0, y_0) .

- As was the case in examples above we can rewrite the equation of the astroid near (x_0, y_0) in the form $y = f(x)$, with an explicit $f(x)$, by solving the equation $x^{2/3} + y^{2/3} = 1$. But again, it is computationally cleaner, and hence less vulnerable to mechanical errors, to use implicit differentiation. So that's what we'll do.
- First up, since (x_0, y_0) lies on the curve, it satisfies

$$x_0^{2/3} + y_0^{2/3} = 1.$$

⁴² Here is where the astroid comes from. Imagine two circles, one of radius $1/4$ and one of radius 1 . Paint a red dot on the smaller circle. Then imagine the smaller circle rolling around the inside of the larger circle. The curve traced by the red dot is our astroid. Google “astroid” (be careful about the spelling) to find animations showing this.

The astroid was first discussed by Johann Bernoulli in 1691–92. It also appears in the work of Leibniz.

- Now, no pretending that $y = f(x)$, this time — just make sure we remember when we differentiate that y changes with x .

$$x^{2/3} + y^{2/3} = 1 \quad \text{start with the curve, and differentiate}$$

$$\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}y' = 0$$

- Note the derivative of $x^{2/3}$, namely $\frac{2}{3}x^{-1/3}$, and the derivative of $y^{2/3}$, namely $\frac{2}{3}y^{-1/3}y'$, are defined only when $x \neq 0$ and $y \neq 0$. We are interested in the case that $x = x_0$ and $y = y_0$. So we better assume that $x_0 \neq 0$ and $y_0 \neq 0$. Probably something weird happens when $x_0 = 0$ or $y_0 = 0$. We'll come back to this shortly.
- To continue on, we set $x = x_0, y = y_0$ in the equation above, and then solve for y' :

$$\frac{2}{3}x_0^{-1/3} + \frac{2}{3}y_0^{-1/3}y'(x) = 0 \implies y'(x_0) = -\left(\frac{y_0}{x_0}\right)^{1/3}$$

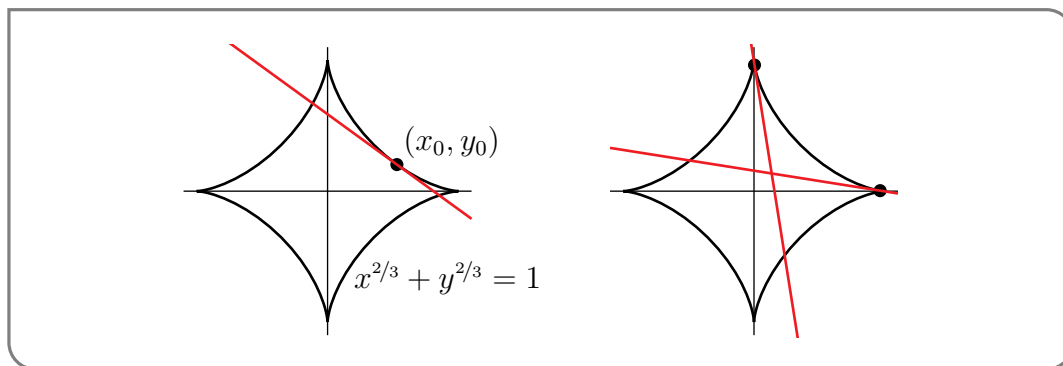
This is the slope of the tangent line and its equation is

$$y = y_0 + f'(x_0)(x - x_0) = y_0 - \left(\frac{y_0}{x_0}\right)^{1/3} (x - x_0)$$

Now let's think a little bit about what the tangent line slope of $-\sqrt[3]{y_0/x_0}$ tells us about the astroid.

- First, as a preliminary observation, note that since $x_0^{2/3} \geq 0$ and $y_0^{2/3} \geq 0$ the equation $x_0^{2/3} + y_0^{2/3} = 1$ of the astroid forces $0 \leq x_0^{2/3}, y_0^{2/3} \leq 1$ and hence $-1 \leq x_0, y_0 \leq 1$.
- For all $x_0, y_0 > 0$ the slope $-\sqrt[3]{y_0/x_0} < 0$. So at all points on the astroid that are in the first quadrant, the tangent line has negative slope, i.e. is “leaning backwards”.
- As x_0 tends to zero, y_0 tends to ± 1 and the tangent line slope tends to infinity. So at points on the astroid near $(0, \pm 1)$, the tangent line is almost vertical.
- As y_0 tends to zero, x_0 tends to ± 1 and the tangent line slope tends to zero. So at points on the astroid near $(\pm 1, 0)$, the tangent line is almost horizontal.

Here is a figure illustrating all this.



Sure enough, as we speculated earlier, something weird does happen to the astroid when x_0 or y_0 is zero. The astroid is pointy, and does not have a tangent there.

Example 2.11.5

2.12 ▲ Inverse Trigonometric Functions

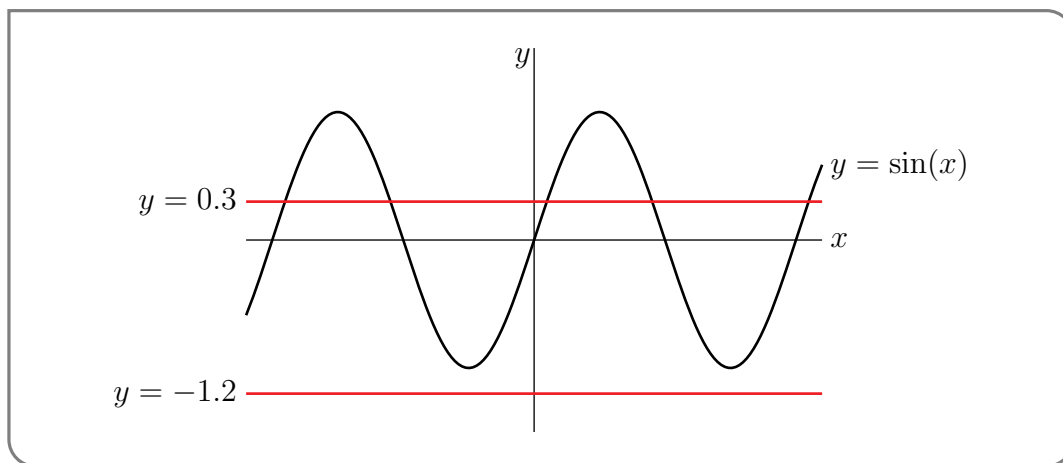
One very useful application of implicit differentiation is to find the derivatives of inverse functions. We have already used this approach to find the derivative of the inverse of the exponential function — the logarithm.

We are now going to consider the problem of finding the derivatives of the inverses of trigonometric functions. Now is a very good time to go back and reread Section 0.6 on inverse functions — especially Definition 0.6.3. Most importantly, given a function $f(x)$, its inverse function $f^{-1}(x)$ only exists, with domain D , when $f(x)$ passes the “horizontal line test”, which says that for each Y in D the horizontal line $y = Y$ intersects the graph $y = f(x)$ exactly once. (That is, $f(x)$ is a one-to-one function.)

Let us start by playing with the sine function and determine how to restrict the domain of $\sin x$ so that its inverse function exists.

Example 2.12.1

Let $y = f(x) = \sin(x)$. We would like to find the inverse function which takes y and returns to us a unique x -value so that $\sin(x) = y$.



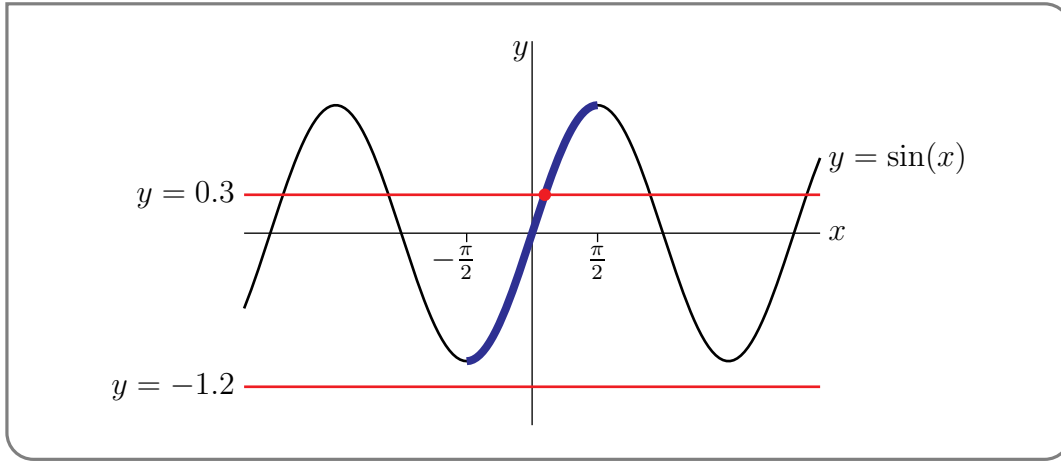
- For each real number Y , the number of x -values that obey $\sin(x) = Y$, is exactly the number of times the horizontal straight line $y = Y$ intersects the graph of $\sin(x)$.
- When $-1 \leq Y \leq 1$, the horizontal line intersects the graph infinitely many times. This is illustrated in the figure above by the line $y = 0.3$.
- On the other hand, when $Y < -1$ or $Y > 1$, the line $y = Y$ never intersects the graph of $\sin(x)$. This is illustrated in the figure above by the line $y = -1.2$.

This is exactly the horizontal line test and it shows that the sine function is not one-to-one.

Now consider the function

$$y = \sin(x) \quad \text{with domain } -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$$

This function has the same formula but the domain has been restricted so that, as we'll now show, the horizontal line test is satisfied.



As we saw above when $|Y| > 1$ no x obeys $\sin(x) = Y$ and, for each $-1 \leq Y \leq 1$, the line $y = Y$ (illustrated in the figure above with $y = 0.3$) crosses the curve $y = \sin(x)$ infinitely many times, so that there are infinitely many x 's that obey $f(x) = \sin x = Y$. However exactly one of those crossings (the dot in the figure) has $-\pi/2 \leq x \leq \pi/2$.

That is, for each $-1 \leq Y \leq 1$, there is exactly one x , call it X , that obeys both

$$\sin X = Y \quad \text{and} \quad -\frac{\pi}{2} \leq X \leq \frac{\pi}{2}$$

That unique value, X , is typically denoted $\arcsin(Y)$. That is

$$\sin(\arcsin(Y)) = Y \quad \text{and} \quad -\frac{\pi}{2} \leq \arcsin(Y) \leq \frac{\pi}{2}$$

Renaming $Y \rightarrow x$, the inverse function $\arcsin(x)$ is defined for all $-1 \leq x \leq 1$ and is determined by the equation

$$\sin(\arcsin(x)) = x \quad \text{and} \quad -\frac{\pi}{2} \leq \arcsin(x) \leq \frac{\pi}{2}. \quad (2.12.1)$$

Note that many texts will use $\sin^{-1}(x)$ to denote arcsine, however we will use $\arcsin(x)$ since we feel that it is clearer⁴³; the reader should recognise both.

Example 2.12.1

Example 2.12.2

Since

$$\sin \frac{\pi}{2} = 1 \quad \sin \frac{\pi}{6} = \frac{1}{2}$$

⁴³ The main reason being that people frequently confuse $\sin^{-1}(x)$ with $(\sin(x))^{-1} = \frac{1}{\sin x}$. We feel that prepending the prefix "arc" less likely to lead to such confusion. The notations $\operatorname{asin}(x)$ and $\operatorname{Arcsin}(x)$ are also used.

and $-\pi/2 \leq \pi/6$, $\pi/2 \leq \pi/2$, we have

$$\arcsin 1 = \frac{\pi}{2} \quad \arcsin \frac{1}{2} = \frac{\pi}{6}$$

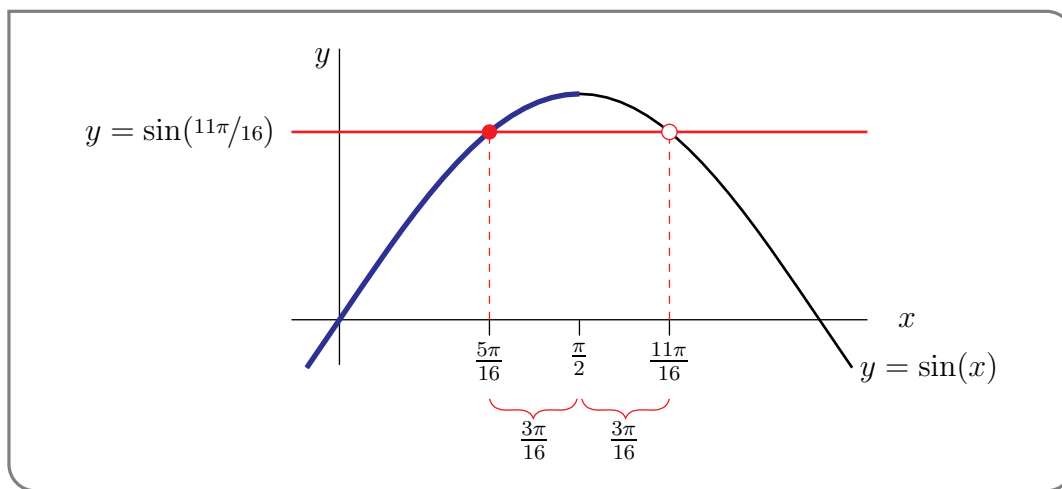
Even though

$$\sin(2\pi) = 0$$

it is **not** true that $\arcsin 0 = 2\pi$, and it is **not** true that $\arcsin(\sin(2\pi)) = 2\pi$, because 2π is not between $-\pi/2$ and $\pi/2$. More generally

$$\begin{aligned} \arcsin(\sin(x)) &= \text{the unique angle } \theta \text{ between } -\pi/2 \text{ and } \pi/2 \text{ obeying } \sin \theta = \sin x \\ &= x \quad \text{if and only if } -\pi/2 \leq x \leq \pi/2 \end{aligned}$$

So, for example, $\arcsin(\sin(11\pi/16))$ cannot be $11\pi/16$ because $11\pi/16$ is bigger than $\pi/2$. So how do we find the correct answer? Start by sketching the graph of $\sin(x)$.



It looks like the graph of $\sin x$ is symmetric about $x = \pi/2$. The mathematical way to say that “the graph of $\sin x$ is symmetric about $x = \pi/2$ ” is “ $\sin(\pi/2 - \theta) = \sin(\pi/2 + \theta)$ ” for all θ . That is indeed true⁴⁴.

Now $11\pi/16 = \pi/2 + 3\pi/16$ so

$$\sin\left(\frac{11\pi}{16}\right) = \sin\left(\frac{\pi}{2} + \frac{3\pi}{16}\right) = \sin\left(\frac{\pi}{2} - \frac{3\pi}{16}\right) = \sin\left(\frac{5\pi}{16}\right)$$

and, since $5\pi/16$ is indeed between $-\pi/2$ and $\pi/2$,

$$\arcsin\left(\sin\left(\frac{11\pi}{16}\right)\right) = \frac{5\pi}{16} \quad \left(\text{and not } \frac{11\pi}{16}\right).$$

Example 2.12.2

44 Indeed both are equal to $\cos \theta$. You can see this by playing with the trig identities in Appendix A.8.

►► Derivatives of Inverse Trig Functions

Now that we have explored the arcsine function we are ready to find its derivative. Let's call

$$\arcsin(x) = \theta(x),$$

so that the derivative we are seeking is $\frac{d\theta}{dx}$. The above equation is (after taking sine of both sides) equivalent to

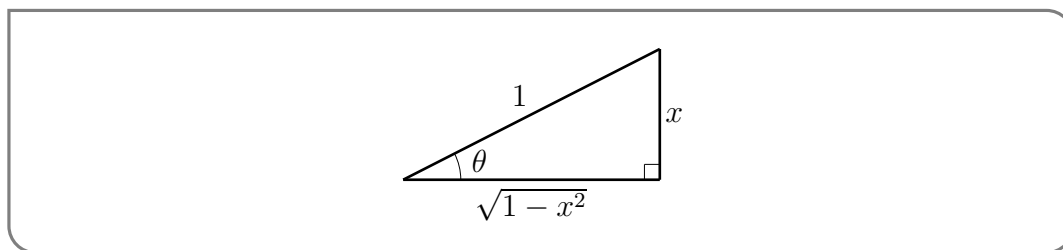
$$\sin(\theta) = x$$

Now differentiate this using implicit differentiation (we just have to remember that θ varies with x and use the chain rule carefully):

$$\begin{aligned} \cos(\theta) \cdot \frac{d\theta}{dx} &= 1 \\ \frac{d\theta}{dx} &= \frac{1}{\cos(\theta)} && \text{substitute } \theta = \arcsin x \\ \frac{d}{dx} \arcsin x &= \frac{1}{\cos(\arcsin x)} \end{aligned}$$

This doesn't look too bad, but it's not really very satisfying because the right hand side is expressed in terms of $\arcsin(x)$ and we do not have an explicit formula for $\arcsin(x)$.

However even without an explicit formula for $\arcsin(x)$, it is a simple matter to get an explicit formula for $\cos(\arcsin(x))$, which is all we need. Just draw a right-angled triangle with one angle being $\arcsin(x)$. This is done in the figure below⁴⁵



Since $\sin(\theta) = x$ (see (2.12.1)), we have made the side opposite the angle θ of length x and the hypotenuse of length 1. Then, by Pythagoras, the side adjacent to θ has length $\sqrt{1 - x^2}$ and so

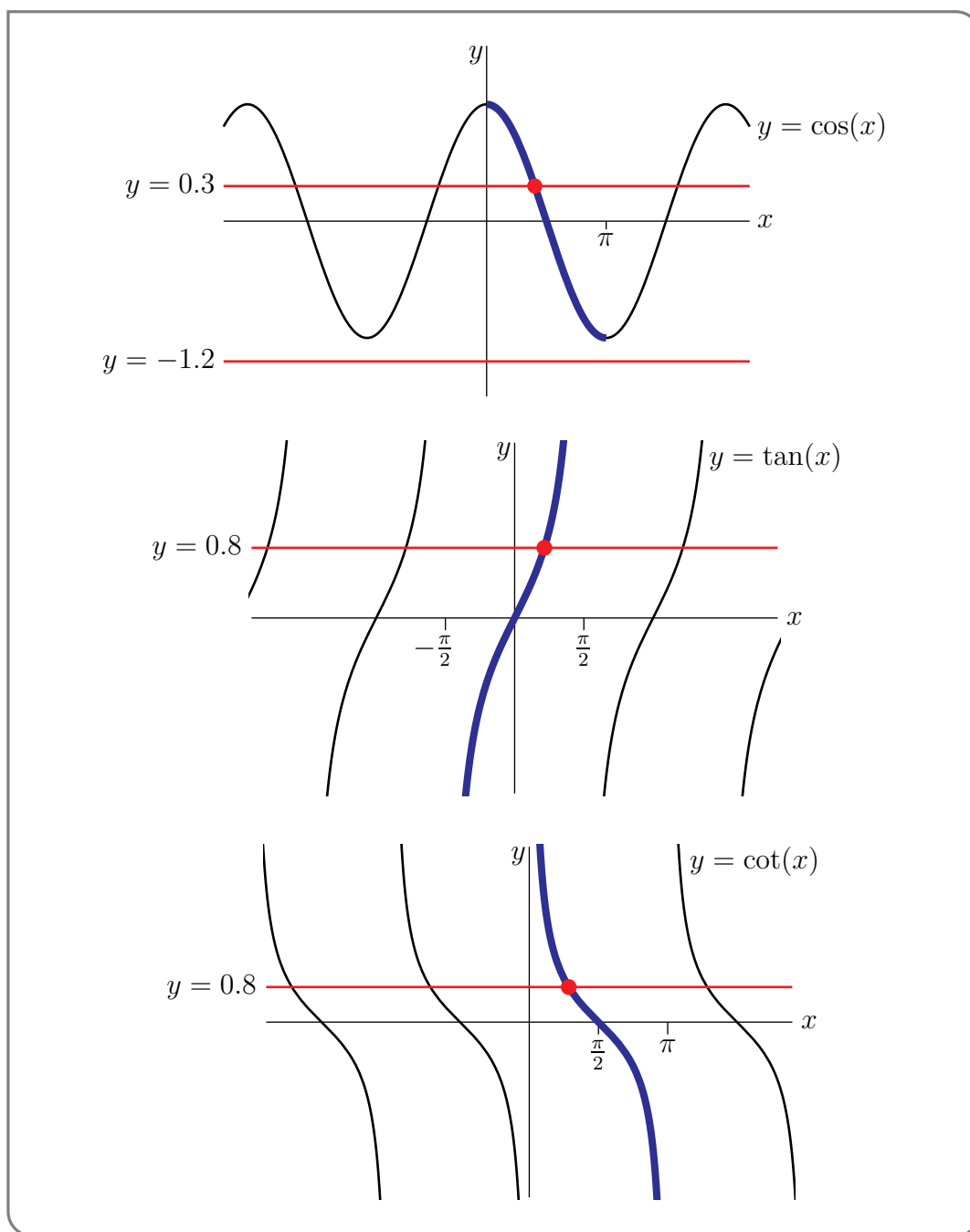
$$\cos(\arcsin(x)) = \cos(\theta) = \sqrt{1 - x^2}$$

which in turn gives us the answer we need:

$$\frac{d}{dx} \arcsin(x) = \frac{1}{\sqrt{1 - x^2}}$$

The definitions for arccos, arctan and arccot are developed in the same way. Here are the graphs that are used.

45 The figure is drawn for the case that $0 \leq \arcsin(x) \leq \pi/2$. Virtually the same argument works for the case $-\pi/2 \leq \arcsin(x) \leq 0$



The definitions for the remaining two inverse trigonometric functions may also be developed in the same way^{46,47}. But it's a little easier to use

$$\csc x = \frac{1}{\sin x} \quad \sec x = \frac{1}{\cos x}$$

- 46 In fact, there are two different widely used definitions of $\operatorname{arcsec} x$. Under our definition, below, $\theta = \operatorname{arcsec} x$ takes values in $0 \leq \theta \leq \pi$. Some people, perfectly legitimately, define $\theta = \operatorname{arcsec} x$ to take values in the union of $0 \leq \theta < \frac{\pi}{2}$ and $\pi \leq \theta < \frac{3\pi}{2}$. Our definition is sometimes called the “trigonometry friendly” definition. The definition itself has the advantage of simplicity. The other definition is sometimes called the “calculus friendly” definition. It eliminates some absolute values and hence simplifies some computations. Similarly, there are two different widely used definitions of $\operatorname{arccsc} x$.
- 47 One could also define $\operatorname{arccot}(x) = \arctan(1/x)$ with $\operatorname{arccot}(0) = \frac{\pi}{2}$. We have chosen not to do so, because the definition we have chosen is both continuous and standard.

Definition 2.12.3.

$\arcsin x$ is defined for $|x| \leq 1$. It is the unique number obeying

$$\sin(\arcsin(x)) = x \quad \text{and} \quad -\frac{\pi}{2} \leq \arcsin(x) \leq \frac{\pi}{2}$$

$\arccos x$ is defined for $|x| \leq 1$. It is the unique number obeying

$$\cos(\arccos(x)) = x \quad \text{and} \quad 0 \leq \arccos(x) \leq \pi$$

$\arctan x$ is defined for all $x \in \mathbb{R}$. It is the unique number obeying

$$\tan(\arctan(x)) = x \quad \text{and} \quad -\frac{\pi}{2} < \arctan(x) < \frac{\pi}{2}$$

$\operatorname{arccsc} x = \arcsin \frac{1}{x}$ is defined for $|x| \geq 1$. It is the unique number obeying

$$\csc(\operatorname{arccsc}(x)) = x \quad \text{and} \quad -\frac{\pi}{2} \leq \operatorname{arccsc}(x) \leq \frac{\pi}{2}$$

Because $\csc(0)$ is undefined, $\operatorname{arccsc}(x)$ never takes the value 0.

$\operatorname{arcsec} x = \arccos \frac{1}{x}$ is defined for $|x| \geq 1$. It is the unique number obeying

$$\sec(\operatorname{arcsec}(x)) = x \quad \text{and} \quad 0 \leq \operatorname{arcsec}(x) \leq \pi$$

Because $\sec(\pi/2)$ is undefined, $\operatorname{arcsec}(x)$ never takes the value $\pi/2$.

$\operatorname{arccot} x$ is defined for all $x \in \mathbb{R}$. It is the unique number obeying

$$\cot(\operatorname{arccot}(x)) = x \quad \text{and} \quad 0 < \operatorname{arccot}(x) < \pi$$

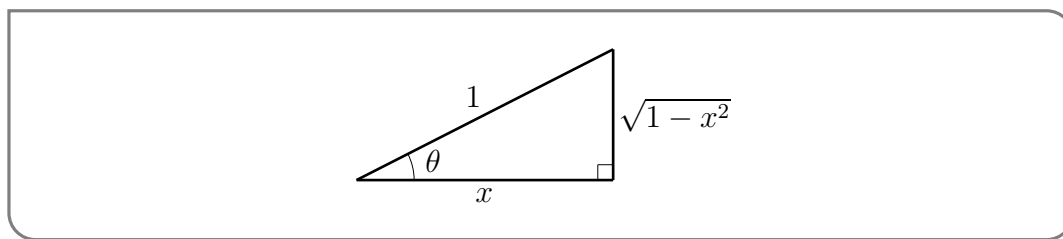
Example 2.12.4

To find the derivative of $\arccos$ we can follow the same steps:

- Write $\arccos(x) = \theta(x)$ so that $\cos \theta = x$ and the desired derivative is $\frac{d\theta}{dx}$.
- Differentiate implicitly, remembering that θ is a function of x :

$$\begin{aligned} -\sin \theta \frac{d\theta}{dx} &= 1 \\ \frac{d\theta}{dx} &= -\frac{1}{\sin \theta} \\ \frac{d}{dx} \arccos x &= -\frac{1}{\sin(\arccos x)}. \end{aligned}$$

- To simplify this expression, again draw the relevant triangle



from which we see

$$\sin(\arccos x) = \sin \theta = \sqrt{1 - x^2}.$$

- Thus

$$\frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1 - x^2}}.$$

Example 2.12.4

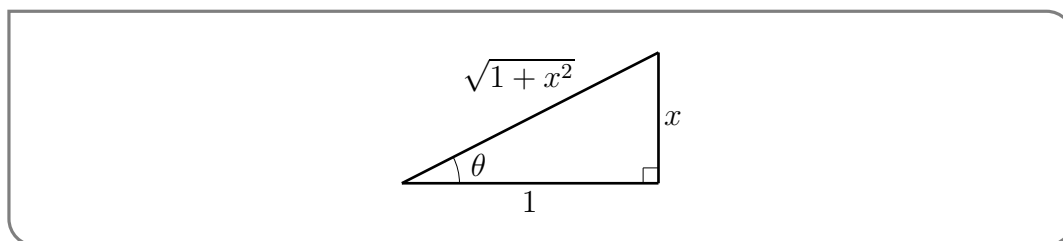
Example 2.12.5

Very similar steps give the derivative of $\arctan x$:

- Start with $\theta = \arctan x$, so $\tan \theta = x$.
- Differentiate implicitly:

$$\begin{aligned} \sec^2 \theta \frac{d\theta}{dx} &= 1 \\ \frac{d\theta}{dx} &= \frac{1}{\sec^2 \theta} = \cos^2 \theta \\ \frac{d}{dx} \arctan x &= \cos^2(\arctan x). \end{aligned}$$

- To simplify this expression, we draw the relevant triangle



from which we see

$$\cos^2(\arctan x) = \cos^2 \theta = \frac{1}{1 + x^2}$$

- Thus

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}.$$

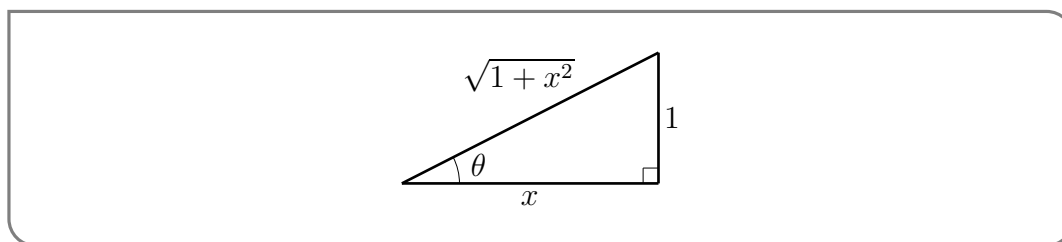
An almost identical computation gives the derivative of $\operatorname{arccot} x$:

- Start with $\theta = \operatorname{arccot} x$, so $\cot \theta = x$.
- Differentiate implicitly:

$$-\csc^2 \theta \frac{d\theta}{dx} = 1$$

$$\frac{d}{dx} \operatorname{arccot} x = \frac{d\theta}{dx} = -\frac{1}{\csc^2 \theta} = -\sin^2 \theta = -\frac{1}{1+x^2}$$

from the triangle



Example 2.12.5

Example 2.12.6

To find the derivative of $\operatorname{arccsc} x$ we can use its definition and the chain rule.

$$\theta = \operatorname{arccsc} x$$

$$\csc \theta = x$$

$$\sin \theta = \frac{1}{x}$$

$$\theta = \arcsin \left(\frac{1}{x} \right)$$

take cosecant of both sides

but $\csc \theta = \frac{1}{\sin \theta}$, so flip both sides

now take arcsine of both sides

Now just differentiate:

$$\begin{aligned} \frac{d\theta}{dx} &= \frac{d}{dx} \arcsin \left(\frac{1}{x} \right) \\ &= \frac{1}{\sqrt{1-x^{-2}}} \cdot \frac{-1}{x^2} \end{aligned}$$

chain rule carefully

Q[34]: Evaluate the derivative of $f(x) = \sqrt[3]{\frac{e^{\csc x^2}}{\sqrt{x^3 - 9} \tan x}}$. You do not have to simplify your answer.

Q[35]: Suppose a particle is moving in the Cartesian plane over time. For any real number $t \geq 0$, the coordinate of the particle at time t is given by $(\sin t, \cos^2 t)$.

- (a) Sketch a graph of the curve traced by the particle in the plane by plotting points, and describe how the particle moves along it over time.
- (b) What is the slope of the curve traced by the particle at time $t = \frac{10\pi}{3}$?

2.10▲ The natural logarithm

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#). Reminder: in these notes, we use $\log x$ to mean $\log_e x$, which is also commonly written elsewhere as $\ln x$.

►► Stage 1

Q[1]: The volume in decibels (dB) of a sound is given by the formula:

$$V(P) = 10 \log_{10} \left(\frac{P}{S} \right)$$

where P is the intensity of the sound and S is the intensity of a standard baseline sound. (That is: S is some constant.)

How much noise will ten speakers make, if each speaker produces 3dB of noise? What about one hundred speakers?

Q[2]: An investment of \$1000 with an interest rate of 5% per year grows to

$$A(t) = 1000e^{t/20}$$

dollars after t years. When will the investment double?

Q[3]: Which of the following expressions, if any, is equivalent to $\log(\cos^2 x)$?

- (a) $2 \log(\cos x)$ (b) $2 \log |\cos x|$ (c) $\log^2(\cos x)$ (d) $\log(\cos x^2)$

►► Stage 2

Q[4]: Differentiate $f(x) = \log(10x)$.

Q[5]: Differentiate $f(x) = \log(x^2)$.

Q[6]: Differentiate $f(x) = \log(x^2 + x)$.

Q[7]: Differentiate $f(x) = \log_{10} x$.

Q[8](*) : Find the derivative of $y = \frac{\log x}{x^3}$.

Q[9]: Evaluate $\frac{d}{d\theta} \log(\sec \theta)$.

Q[10]: Differentiate the function $f(x) = e^{\cos(\log x)}$.

Q[11](*) : Evaluate the derivative. You do not need to simplify your answer.

$$y = \log(x^2 + \sqrt{x^4 + 1})$$

Q[12](*) : Differentiate $\sqrt{-\log(\cos x)}$.

Q[13](*) : Calculate and simplify the derivative of $\log(x + \sqrt{x^2 + 4})$.

Q[14](*) : Evaluate the derivative of $g(x) = \log(e^{x^2} + \sqrt{1 + x^4})$.

Q[15](*) : Evaluate the derivative of the following function at $x = 1$: $g(x) = \log\left(\frac{2x-1}{2x+1}\right)$.

Q[16]: Evaluate the derivative of the function $f(x) = \log\left(\sqrt{\frac{(x^2+5)^3}{x^4+10}}\right)$.

Q[17]: Evaluate $f'(2)$ if $f(x) = \log(g(xh(x)))$, $h(2) = 2$, $h'(2) = 3$, $g(4) = 3$, $g'(4) = 5$.

Q[18](*) : Differentiate the function

$$g(x) = \pi^x + x^\pi.$$

Q[19]: Differentiate $f(x) = x^x$.

Q[20](*) : Find $f'(x)$ if $f(x) = x^x + \log_{10} x$.

Q[21]: Differentiate $f(x) = \sqrt[4]{\frac{(x^4+12)(x^4-x^2+2)}{x^3}}$.

Q[22]: Differentiate $f(x) = (x+1)(x^2+1)^2(x^3+1)^3(x^4+1)^4(x^5+1)^5$.

Q[23]: Differentiate $f(x) = \left(\frac{5x^2 + 10x + 15}{3x^4 + 4x^3 + 5} \right) \left(\frac{1}{10(x+1)} \right)$.

Q[24](*) : Let $f(x) = (\cos x)^{\sin x}$, with domain $0 < x < \frac{\pi}{2}$. Find $f'(x)$.

Q[25](*) : Find the derivative of $(\tan(x))^x$, when x is in the interval $(0, \pi/2)$.

Q[26](*) : Find $f'(x)$ if $f(x) = (x^2 + 1)^{(x^2+1)}$

Q[27](*) : Differentiate $f(x) = (x^2 + 1)^{\sin(x)}$.

Q[28](*) : Let $f(x) = x^{\cos^3(x)}$, with domain $(0, \infty)$. Find $f'(x)$.

Q[29](*) : Differentiate $f(x) = (3 + \sin(x))^{x^2-3}$.

► Stage 3

Q[30]: Let $f(x)$ and $g(x)$ be differentiable functions, with $f(x) > 0$. Evaluate $\frac{d}{dx} \{ [f(x)]^{g(x)} \}$.

Q[31]: Let $f(x)$ be a function whose range includes only positive numbers. Show that the curves $y = f(x)$ and $y = \log(f(x))$ have horizontal tangent lines at the same values of x .

2.11▲ Implicit Differentiation

Exercises

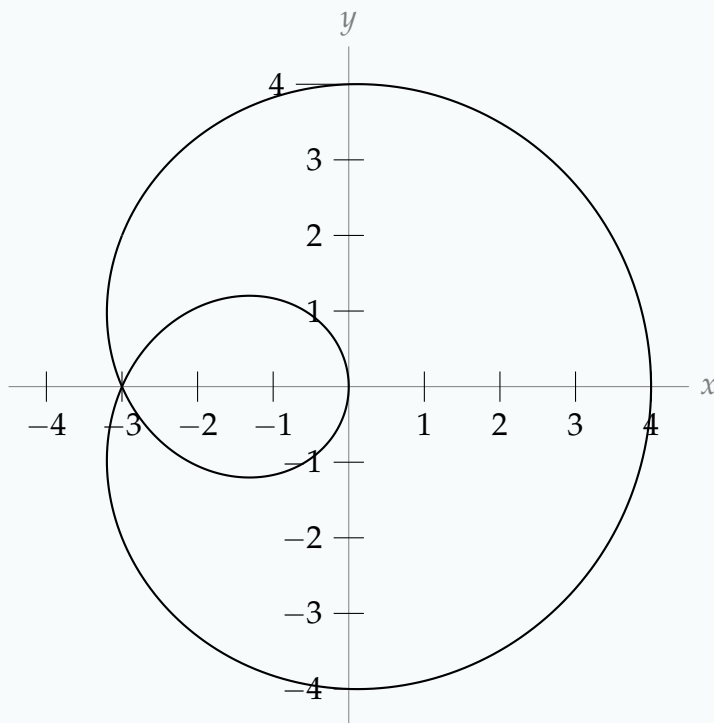
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► Stage 1

Q[1]: If we implicitly differentiate $x^2 + y^2 = 1$, we get the equation $2x + 2yy' = 0$. In the step where we differentiate y^2 to obtain $2yy'$, which rule(s) below are we using?

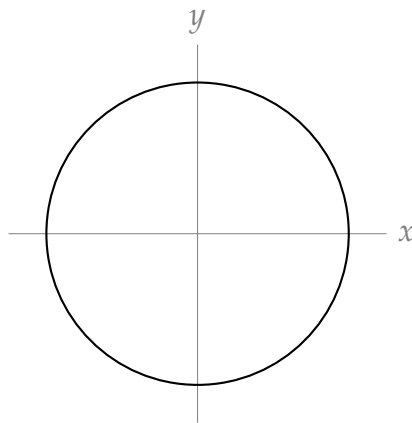
- (a) power rule (b) chain rule (c) quotient rule
(d) derivatives of exponential functions

Q[2]: Using the picture below, estimate $\frac{dy}{dx}$ at the three points where the curve crosses the y -axis.



Remark: for this curve, one value of x may correspond to multiple values of y . So, we cannot express this curve as $y = f(x)$ for any function x . This is one typical situation where we might use implicit differentiation.

Q[3]: Consider the unit circle, formed by all points (x, y) that satisfy $x^2 + y^2 = 1$.



- Is there a function $f(x)$ so that $y = f(x)$ completely describes the unit circle? That is, so that the points (x, y) that make the equation $y = f(x)$ true are exactly the same points that make the equation $x^2 + y^2 = 1$ true?
- Is there a function $f'(x)$ so that $y = f'(x)$ completely describes the slope of the unit circle? That is, so that for every point (x, y) on the unit circle, the slope of the tangent line to the circle at that point is given by $f'(x)$?

- (c) Use implicit differentiation to find an expression for $\frac{dy}{dx}$. Simplify until the expression is a function in terms of x only (not y), or explain why this is impossible.

►► Stage 2

Q[4](*) : Find $\frac{dy}{dx}$ if $xy + e^x + e^y = 1$.

Q[5](*) : If $e^y = xy^2 + x$, compute $\frac{dy}{dx}$.

Q[6](*) : If $x^2 \tan(\pi y/4) + 2x \log(y) = 16$, then find y' at the points where $y = 1$.

Q[7](*) : If $x^3 + y^4 = \cos(x^2 + y)$ compute $\frac{dy}{dx}$.

Q[8](*) : If $x^2 e^y + 4x \cos(y) = 5$, then find y' at the points where $y = 0$.

Q[9](*) : If $x^2 + y^2 = \sin(x + y)$ compute $\frac{dy}{dx}$.

Q[10](*) : If $x^2 \cos(y) + 2xe^y = 8$, then find y' at the points where $y = 0$.

Q[11] : At what points on the ellipse $x^2 + 3y^2 = 1$ is the tangent line parallel to the line $y = x$?

Q[12](*) : For the curve defined by the equation $\sqrt{xy} = x^2y - 2$, find the slope of the tangent line at the point $(1, 4)$.

Q[13](*) : If $x^2y^2 + x \sin(y) = 4$, find $\frac{dy}{dx}$.

►► Stage 3

Q[14](*) : If $x^2 + (y + 1)e^y = 5$, then find y' at the points where $y = 0$.

Q[15] : For what values of x do the circle $x^2 + y^2 = 1$ and the ellipse $x^2 + 3y^2 = 1$ have parallel tangent lines?

2.12▲ Inverse Trigonometric Functions

Exercises

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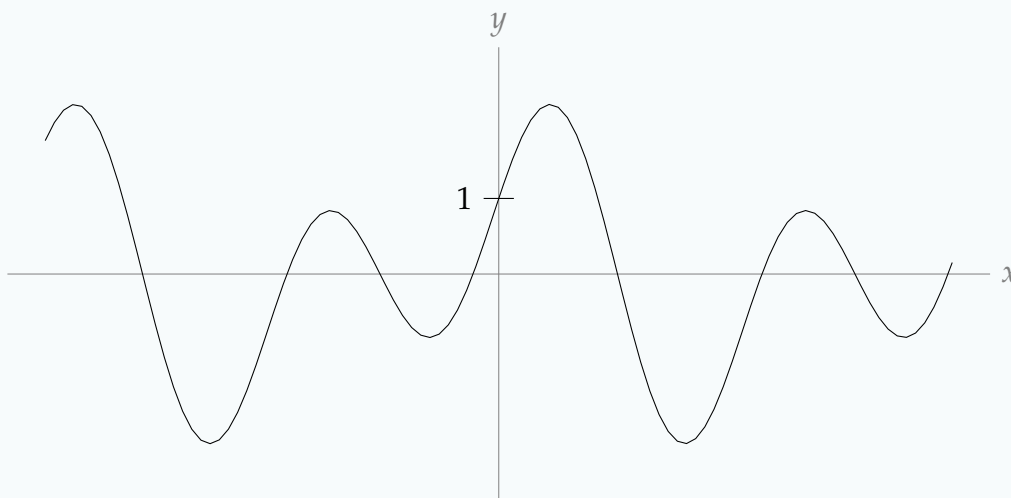
► Stage 1

Q[1]: Give the domains of each of the following functions.

(a) $f(x) = \arcsin(\cos x)$ (b) $g(x) = \operatorname{arccsc}(\cos x)$ (c) $h(x) = \sin(\arccos x)$

Q[2]: A particle starts moving at time $t = 10$, and it bobs up and down, so that its height at time $t \geq 10$ is given by $\cos t$. True or false: the particle has height 1 at time $t = \arccos(1)$.

Q[3]: The curve $y = f(x)$ is shown below, for some function f . Restrict f to the largest possible interval containing 0 over which it is one-to-one, and sketch the curve $y = f^{-1}(x)$.



Q[4]: Let a be some constant. Where does the curve $y = ax + \cos x$ have a horizontal tangent line?

Q[5]: Define a function $f(x) = \arcsin x + \operatorname{arccsc} x$. What is the domain of $f(x)$? Where is $f(x)$ differentiable?

► Stage 2

Q[6]: Differentiate $f(x) = \arcsin\left(\frac{x}{3}\right)$. What is the domain of $f(x)$?

Q[7]: Differentiate $f(t) = \frac{\arccos t}{t^2 - 1}$. What is the domain of $f(t)$?

Q[8]: Differentiate $f(x) = \operatorname{arcsec}(-x^2 - 2)$. What is the domain of $f(x)$?

Q[9]: Differentiate $f(x) = \frac{1}{a} \arctan\left(\frac{x}{a}\right)$, where a is a nonzero constant. What is the domain of $f(x)$?

Q[10]: Differentiate $f(x) = x \arcsin x + \sqrt{1 - x^2}$. What is the domain of $f(x)$?

Q[11]: For which values of x is the tangent line to $y = \arctan(x^2)$ horizontal?

Q[12]: Evaluate $\frac{d}{dx}\{\arcsin x + \arccos x\}$.

Q[13](*) : Find the derivative of $y = \arcsin\left(\frac{1}{x}\right)$.

Q[14](*) : Find the derivative of $y = \arctan\left(\frac{1}{x}\right)$.

Q[15](*) : Calculate and simplify the derivative of $(1 + x^2) \arctan x$.

Q[16]: Show that $\frac{d}{dx}\{\sin(\arctan(x))\} = (x^2 + 1)^{-3/2}$.

Q[17]: Show that $\frac{d}{dx}\{\cot(\arcsin(x))\} = \frac{-1}{x^2\sqrt{1-x^2}}$.

Q[18](*) : Determine all points on the curve $y = \arcsin x$ where the tangent line is parallel to the line $y = 2x + 9$.

Q[19]: For which values of x does the function $f(x) = \arctan(\csc x)$ have a horizontal tangent line?

► Stage 3

Q[20](*) : Let $f(x) = x + \cos x$, and let $g(y) = f^{-1}(y)$ be the inverse function. Determine $g'(y)$.

Q[21](*) : $f(x) = 2x - \sin(x)$ is one-to-one. Find $(f^{-1})'(\pi - 1)$.

Q[22](*) : $f(x) = e^x + x$ is one-to-one. Find $(f^{-1})'(e + 1)$.

Q[23]: Differentiate $f(x) = [\sin x + 2]^{\operatorname{arccsc} x}$. What is the domain of this function?

Q[24]: Suppose you can't remember whether the derivative of arcsine is $\frac{1}{\sqrt{1-x^2}}$ or $\frac{1}{\sqrt{x^2-1}}$. Describe how the domain of arcsine suggests that one of these is wrong.

Q[25]: Evaluate $\lim_{x \rightarrow 1} \left((x-1)^{-1} \left(\arctan x - \frac{\pi}{4} \right) \right)$.

Q[26]: Suppose $f(2x+1) = \frac{5x-9}{3x+7}$. Evaluate $f^{-1}(7)$.

Remark: The standard way to write this problem is to omit the notation $f(x)$, and let the variable y stand for two functions. When t is the variable, $y(t) = \cos^2 t$ gives the y -coordinate of the particle at time t . When x is the variable, $y(x)$ gives the y -coordinate of the particle given its position along the x -axis. This is an abuse of notation, because if we write $y(1)$, it is not clear whether we are referring to the y -coordinate of the particle when $t = 1$ (in this case, $y = \cos^2(1) \approx 0.3$), or the y -coordinate of the particle when $x = 1$ (in this case, looking at our table of values, $y = 0$). Although this notation is not strictly “correct,” it is very commonly used. So, you might see a solution that looks like this:

The slope of the curve is $\frac{dy}{dx}$. To find $\frac{dy}{dx}$, we use the chain rule:

$$\begin{aligned}\frac{dy}{dt} &= \frac{dy}{dx} \cdot \frac{dx}{dt} \\ \frac{d}{dt} \{\cos^2 t\} &= \frac{dy}{dx} \cdot \frac{d}{dt} \{\sin t\} \\ -2 \cos t \sin t &= \frac{dy}{dx} \cdot \cos t \\ \frac{dy}{dx} &= -2 \sin t\end{aligned}$$

So, when $t = \frac{10\pi}{3}$,

$$\frac{dy}{dx} = -2 \sin \left(\frac{10\pi}{3} \right) = -2 \left(-\frac{\sqrt{3}}{2} \right) = \sqrt{3}.$$

In this case, it is up to the reader to understand when y is used as a function of t , and when it is used as a function of x . This notation (using y to be two functions, $y(t)$ and $y(x)$) is actually the accepted standard, so you should be able to understand it.

Solutions to Exercises 2.10 — Jump to [TABLE OF CONTENTS](#)

S-1: We are given that one speaker produces 3dB. So if P is the power of one speaker,

$$3 = V(P) = 10 \log_{10} \left(\frac{P}{S} \right).$$

So, for ten speakers:

$$\begin{aligned}V(10P) &= 10 \log_{10} \left(\frac{10P}{S} \right) = 10 \log_{10} \left(\frac{P}{S} \right) + 10 \log_{10} (10) \\ &= 3 + 10(1) = 13\text{dB}\end{aligned}$$

and for one hundred speakers:

$$\begin{aligned}V(100P) &= 10 \log_{10} \left(\frac{100P}{S} \right) = 10 \log_{10} \left(\frac{P}{S} \right) + 10 \log_{10} (100) \\ &= 3 + 10(2) = 23\text{dB}\end{aligned}$$

S-2: The investment doubles when it hits \$2000. So, we find the value of t that gives $A(t) = 2000$:

$$\begin{aligned}2000 &= A(t) \\2000 &= 1000e^{t/20} \\2 &= e^{t/20} \\\log 2 &= \frac{t}{20} \\20 \log 2 &= t\end{aligned}$$

S-3: From our logarithm rules, we know that when y is *positive*, $\log(y^2) = 2 \log y$. However, the expression $\cos x$ does not always take on positive values, so (a) is not correct. (For instance, when $x = \pi$, $\log(\cos^2 x) = \log(\cos^2 \pi) = \log((-1)^2) = \log(1) = 0$, while $2 \log(\cos \pi) = 2 \log(-1)$, which does not exist.)

Because $\cos^2 x$ is never negative, we notice that $\cos^2 x = |\cos x|^2$. When $\cos x$ is nonzero, $|\cos x|$ is positive, so our logarithm rules tell us $\log(|\cos x|^2) = 2 \log |\cos x|$. When $\cos x$ is exactly zero, then both $\log(\cos^2 x)$ and $2 \log |\cos x|$ do not exist. So, $\log(\cos^2 x) = 2 \log |\cos x|$.

S-4:

- Solution 1: Using the chain rule, $\frac{d}{dx} \{\log(10x)\} = \frac{1}{10x} \cdot 10 = \frac{1}{x}$.
- Solution 2: Simplifying, $\frac{d}{dx} \{\log(10x)\} = \frac{d}{dx} \{\log(10) + \log x\} = 0 + \frac{1}{x} = \frac{1}{x}$.

S-5:

- Solution 1: Using the chain rule, $\frac{d}{dx} \{\log(x^2)\} = \frac{1}{x^2} \cdot 2x = \frac{2}{x}$.
- Solution 2: Simplifying, $\frac{d}{dx} \{\log(x^2)\} = \frac{d}{dx} \{2 \log(x)\} = \frac{2}{x}$.

S-6: Don't be fooled by a common mistake: $\log(x^2 + x)$ is *not* the same as $\log(x^2) + \log x$. We differentiate using the chain rule: $\frac{d}{dx} \{\log(x^2 + x)\} = \frac{1}{x^2 + x} \cdot (2x + 1) = \frac{2x + 1}{x^2 + x}$.

S-7: We know the derivative of the natural logarithm (base e), so we use the base-change formula:

$$f(x) = \log_{10} x = \frac{\log x}{\log 10}$$

Since $\log 10$ is a constant:

$$f'(x) = \frac{1}{x \log 10}.$$

S-8:

- Solution 1: Using the quotient rule,

$$y' = \frac{x^3 \frac{1}{x} - (\log x) \cdot 3x^2}{x^6} = \frac{x^2 - 3x^2 \log x}{x^6} = \frac{1 - 3 \log x}{x^4}.$$

- Solution 2: Using the product rule with $y = \log x \cdot x^{-3}$,

$$y' = \frac{1}{x} x^{-3} + \log x \cdot (-3)x^{-4} = x^{-4}(1 - 3 \log x)$$

S-9: Using the chain rule,

$$\begin{aligned} \frac{d}{d\theta} \log(\sec \theta) &= \frac{1}{\sec \theta} \cdot (\sec \theta \cdot \tan \theta) \\ &= \tan \theta \end{aligned}$$

Remark: the domain of the function $\log(\sec \theta)$ is those values of θ for which $\sec \theta$ is positive: so, the intervals $\left(\left(2n - \frac{1}{2}\right) \pi, \left(2n + \frac{1}{2}\right) \pi \right)$ where n is any integer. Certainly the tangent function has a larger domain than this, but outside the domain of $\log(\sec \theta)$, $\tan \theta$ is not the derivative of $\log(\sec \theta)$.

S-10: Let's start in with the chain rule.

$$f'(x) = e^{\cos(\log x)} \cdot \frac{d}{dx} \{ \cos(\log x) \}$$

We'll need the chain rule again:

$$\begin{aligned} &= e^{\cos(\log x)} (-\sin(\log x)) \cdot \frac{d}{dx} \{ \log x \} \\ &= e^{\cos(\log x)} (-\sin(\log x)) \cdot \frac{1}{x} \\ &= \frac{-e^{\cos(\log x)} \sin(\log x)}{x} \end{aligned}$$

Remark: Although we have a logarithm in the exponent, we can't cancel. The expression $e^{\cos(\log x)}$ is *not* the same as the expression $x^{\cos x}$, or $\cos x$.

S-11:

$$y = \log(x^2 + \sqrt{x^4 + 1})$$

So, we'll need the chain rule:

$$\begin{aligned} y' &= \frac{\frac{d}{dx} \{x^2 + \sqrt{x^4 + 1}\}}{x^2 + \sqrt{x^4 + 1}} \\ &= \frac{2x + \frac{d}{dx} \{\sqrt{x^4 + 1}\}}{x^2 + \sqrt{x^4 + 1}} \end{aligned}$$

We need the chain rule again:

$$\begin{aligned} &= \frac{2x + \frac{\frac{d}{dx} \{x^4 + 1\}}{2\sqrt{x^4 + 1}}}{x^2 + \sqrt{x^4 + 1}} \\ &= \frac{2x + \frac{4x^3}{2\sqrt{x^4 + 1}}}{x^2 + \sqrt{x^4 + 1}}. \end{aligned}$$

S-12: This requires us to apply the chain rule twice.

$$\begin{aligned} \frac{d}{dx} \left\{ \sqrt{-\log(\cos x)} \right\} &= \frac{1}{2\sqrt{-\log(\cos x)}} \cdot \frac{d}{dx} \{-\log(\cos x)\} \\ &= -\frac{1}{2\sqrt{-\log(\cos x)}} \cdot \frac{1}{\cos x} \frac{d}{dx} \{\cos x\} \\ &= -\frac{1}{2\sqrt{-\log(\cos x)}} \cdot \frac{1}{\cos x} \cdot (-\sin x) \\ &= \frac{\tan x}{2\sqrt{-\log(\cos x)}} \end{aligned}$$

Remark: it looks strange to see a negative sign in the argument of a square root. Since the cosine function always gives values that are at most 1, $\log(\cos x)$ is always negative or zero over its domain. So, $\sqrt{\log(\cos x)}$ is only defined for the points where $\cos x = 1$ (and so $\log(\cos x) = 0$ —this isn't a very interesting function! In contrast, $-\log(\cos x)$ is always positive or zero over its domain – and therefore we can always take its square root.

S-13: Under the chain rule, $\frac{d}{dx} \log f(x) = \frac{1}{f(x)} f'(x)$. So

$$\begin{aligned} \frac{d}{dx} \left\{ \log(x + \sqrt{x^2 + 4}) \right\} &= \frac{1}{x + \sqrt{x^2 + 4}} \cdot \frac{d}{dx} \{x + \sqrt{x^2 + 4}\} \\ &= \frac{1}{x + \sqrt{x^2 + 4}} \cdot \left(1 + \frac{2x}{2\sqrt{x^2 + 4}} \right) \\ &= \frac{1}{x + \sqrt{x^2 + 4}} \cdot \left(\frac{2\sqrt{x^2 + 4} + 2x}{2\sqrt{x^2 + 4}} \right) \\ &= \frac{1}{\sqrt{x^2 + 4}} \end{aligned}$$

S-14: Using the chain rule,

$$\begin{aligned}g'(x) &= \frac{\frac{d}{dx}\{e^{x^2} + \sqrt{1+x^4}\}}{e^{x^2} + \sqrt{1+x^4}} \\&= \frac{2xe^{x^2} + \frac{4x^3}{2\sqrt{1+x^4}}}{e^{x^2} + \sqrt{1+x^4}} \left(\frac{\sqrt{1+x^4}}{\sqrt{1+x^4}} \right) \\&= \frac{2xe^{x^2}\sqrt{1+x^4} + 2x^3}{e^{x^2}\sqrt{1+x^4} + 1+x^4}\end{aligned}$$

S-15: Using logarithm rules makes this an easier problem:

$$\begin{aligned}g(x) &= \log(2x-1) - \log(2x+1) \\ \text{So, } g'(x) &= \frac{2}{2x-1} - \frac{2}{2x+1} \\ \text{and } g'(1) &= \frac{2}{1} - \frac{2}{3} = \frac{4}{3}\end{aligned}$$

S-16: We begin by simplifying:

$$\begin{aligned}f(x) &= \log \left(\sqrt{\frac{(x^2+5)^3}{x^4+10}} \right) \\&= \log \left(\left(\frac{(x^2+5)^3}{x^4+10} \right)^{1/2} \right) \\&= \frac{1}{2} \log \left(\frac{(x^2+5)^3}{x^4+10} \right) \\&= \frac{1}{2} \left[\log((x^2+5)^3) - \log(x^4+10) \right] \\&= \frac{1}{2} \left[3 \log(x^2+5) - \log(x^4+10) \right]\end{aligned}$$

Now, we differentiate using the chain rule:

$$\begin{aligned}f'(x) &= \frac{1}{2} \left[3 \frac{2x}{x^2+5} - \frac{4x^3}{x^4+10} \right] \\&= \frac{3x}{x^2+5} - \frac{2x^3}{x^4+10}\end{aligned}$$

Remark: it is a common mistake to write $\log(x^2+4)$ as $\log(x^2) + \log(4)$. These expressions are not equivalent!

S-17: We use the chain rule twice, followed by the product rule:

$$\begin{aligned}f'(x) &= \frac{1}{g(xh(x))} \cdot \frac{d}{dx}\{g(xh(x))\} \\&= \frac{1}{g(xh(x))} \cdot g'(xh(x)) \cdot \frac{d}{dx}\{xh(x)\} \\&= \frac{1}{g(xh(x))} \cdot g'(xh(x)) \cdot [h(x) + xh'(x)]\end{aligned}$$

In particular, when $x = 2$:

$$\begin{aligned}f'(2) &= \frac{1}{g(2h(2))} \cdot g'(2h(2)) \cdot [h(2) + 2h'(2)] \\&= \frac{g'(4)}{g(4)} [2 + 2 \times 3] = \frac{5}{3} [2 + 2 \times 3] \\&= \frac{40}{3}\end{aligned}$$

S-18: In the text, we saw that $\frac{d}{dx}\{a^x\} = a^x \log a$ for any constant a . So,

$$\frac{d}{dx}\{\pi^x\} = \pi^x \log \pi.$$

By the power rule, $\frac{d}{dx}\{x^\pi\} = \pi x^{\pi-1}$.

Therefore, $g'(x) = \pi^x \log \pi + \pi x^{\pi-1}$.

Remark: we had to use two different rules for the two different terms in $g(x)$. Although the functions π^x and x^π look superficially the same, they behave differently, as do their derivatives. A function of the form $(\text{constant})^x$ is an exponential function and *not eligible for the power rule*, while a function of the form x^{constant} is exactly the class of function the power rule applies to.

S-19: We have the power rule to tell us the derivative of functions of the form x^n , where n is a constant. However, here our exponent is not a constant. Similarly, in this section we learned the derivative of functions of the form a^x , where a is a constant, but again, our base is not a constant! Although the result $\frac{d}{dx}a^x = a^x \log a$ is not what we need, the *method* used to differentiate a^x will tell us the derivative of x^x .

We'll set $g(x) = \log(x^x)$, because now we can use logarithm rules to simplify:

$$g(x) = \log(f(x)) = x \log x$$

Now, we can use the product rule to differentiate the right side, and the chain rule to differentiate $\log(f(x))$:

$$g'(x) = \frac{f'(x)}{f(x)} = \log x + x \frac{1}{x} = \log x + 1$$

Finally, we solve for $f'(x)$:

$$f'(x) = f(x)(\log x + 1) = x^x(\log x + 1)$$

S-20: In Question **19**, we saw $\frac{d}{dx} \{x^x\} = x^x(\log x + 1)$. Using the base-change formula, $\log_{10}(x) = \frac{\log x}{\log 10}$. Since $\log_{10}$ is a constant,

$$\begin{aligned} f'(x) &= \frac{d}{dx} \left\{ x^x + \frac{\log x}{\log 10} \right\} \\ &= x^x(\log x + 1) + \frac{1}{x \log 10} \end{aligned}$$

S-21: Rather than set in with a terrible chain rule problem, we'll use logarithmic differentiation. Instead of differentiating $f(x)$, we differentiate a new function $\log(f(x))$, after simplifying.

$$\begin{aligned} \log(f(x)) &= \log \sqrt[4]{\frac{(x^4 + 12)(x^4 - x^2 + 2)}{x^3}} \\ &= \frac{1}{4} \log \left(\frac{(x^4 + 12)(x^4 - x^2 + 2)}{x^3} \right) \\ &= \frac{1}{4} \left(\log(x^4 + 12) + \log(x^4 - x^2 + 2) - 3 \log x \right) \end{aligned}$$

Now that we've simplified, we can efficiently differentiate both sides. It is important to remember that we aren't differentiating $f(x)$ directly—we're differentiating $\log(f(x))$.

$$\frac{f'(x)}{f(x)} = \frac{1}{4} \left(\frac{4x^3}{x^4 + 12} + \frac{4x^3 - 2x}{x^4 - x^2 + 2} - \frac{3}{x} \right)$$

Our final step is to solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) \frac{1}{4} \left(\frac{4x^3}{x^4 + 12} + \frac{4x^3 - 2x}{x^4 - x^2 + 2} - \frac{3}{x} \right) \\ &= \frac{1}{4} \left(\sqrt[4]{\frac{(x^4 + 12)(x^4 - x^2 + 2)}{x^3}} \right) \left(\frac{4x^3}{x^4 + 12} + \frac{4x^3 - 2x}{x^4 - x^2 + 2} - \frac{3}{x} \right) \end{aligned}$$

It was possible to differentiate this function without logarithms, but the logarithms make it more efficient.

S-22: It's possible to do this using the product rule a number of times, but it's easier to use logarithmic differentiation. Set

$$g(x) = \log(f(x)) = \log \left[(x+1)(x^2+1)^2(x^3+1)^3(x^4+1)^4(x^5+1)^5 \right]$$

Now we can use logarithm rules to change $g(x)$ into a form that is friendlier to differentiate:

$$\begin{aligned} &= \log(x+1) + \log(x^2+1)^2 + \log(x^3+1)^3 + \log(x^4+1)^4 + \log(x^5+1)^5 \\ &= \log(x+1) + 2\log(x^2+1) + 3\log(x^3+1) + 4\log(x^4+1) + 5\log(x^5+1) \end{aligned}$$

Now, we differentiate $g(x)$ using the chain rule:

$$g'(x) = \frac{f'(x)}{f(x)} = \frac{1}{x+1} + \frac{4x}{x^2+1} + \frac{9x^2}{x^3+1} + \frac{16x^3}{x^4+1} + \frac{25x^4}{x^5+1}$$

Finally, we solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) \left[\frac{1}{x+1} + \frac{4x}{x^2+1} + \frac{9x^2}{x^3+1} + \frac{16x^3}{x^4+1} + \frac{25x^4}{x^5+1} \right] \\ &= (x+1)(x^2+1)^2(x^3+1)^3(x^4+1)^4(x^5+1)^5 \\ &\quad \cdot \left[\frac{1}{x+1} + \frac{4x}{x^2+1} + \frac{9x^2}{x^3+1} + \frac{16x^3}{x^4+1} + \frac{25x^4}{x^5+1} \right] \end{aligned}$$

S-23: We could do this with quotient and product rules, but it would be pretty painful. Instead, let's use a logarithm.

$$\begin{aligned} f(x) &= \left(\frac{5x^2 + 10x + 15}{3x^4 + 4x^3 + 5} \right) \left(\frac{1}{10(x+1)} \right) = \left(\frac{x^2 + 2x + 3}{3x^4 + 4x^3 + 5} \right) \left(\frac{1}{2(x+1)} \right) \\ \log(f(x)) &= \log \left[\left(\frac{x^2 + 2x + 3}{3x^4 + 4x^3 + 5} \right) \left(\frac{1}{2(x+1)} \right) \right] \\ &= \log \left(\frac{x^2 + 2x + 3}{3x^4 + 4x^3 + 5} \right) + \log \left(\frac{1}{2(x+1)} \right) \\ &= \log(x^2 + 2x + 3) - \log(3x^4 + 4x^3 + 5) - \log(x+1) - \log(2) \end{aligned}$$

Now we have a function that we can differentiate more cleanly than our original function.

$$\begin{aligned} \frac{d}{dx} \{ \log(f(x)) \} &= \frac{d}{dx} \left\{ \log(x^2 + 2x + 3) - \log(3x^4 + 4x^3 + 5) - \log(x+1) - \log(2) \right\} \\ \frac{f'(x)}{f(x)} &= \frac{2x+2}{x^2+2x+3} - \frac{12x^3+12x^2}{3x^4+4x^3+5} - \frac{1}{x+1} \\ &= \frac{2(x+1)}{x^2+2x+3} - \frac{12x^2(x+1)}{3x^4+4x^3+5} - \frac{1}{x+1} \end{aligned}$$

Finally, we solve for $f(x)$:

$$\begin{aligned} f'(x) &= f(x) \left(\frac{2(x+1)}{x^2+2x+3} - \frac{12x^2(x+1)}{3x^4+4x^3+5} - \frac{1}{x+1} \right) \\ &= \left(\frac{x^2+2x+3}{3x^4+4x^3+5} \right) \left(\frac{1}{2(x+1)} \right) \left(\frac{2(x+1)}{x^2+2x+3} - \frac{12x^2(x+1)}{3x^4+4x^3+5} - \frac{1}{x+1} \right) \\ &= \left(\frac{x^2+2x+3}{3x^4+4x^3+5} \right) \left(\frac{1}{x^2+2x+3} - \frac{6x^2}{3x^4+4x^3+5} - \frac{1}{2(x+1)^2} \right) \end{aligned}$$

S-24: Since $f(x)$ has the form of a function raised to a functional power, we will use logarithmic differentiation.

$$\log(f(x)) = \log((\cos x)^{\sin x}) = \sin x \cdot \log(\cos x)$$

Logarithm rules allowed us to simplify. Now, we differentiate both sides of this equation:

$$\begin{aligned} \frac{f'(x)}{f(x)} &= (\cos x) \log(\cos x) + \sin x \cdot \frac{-\sin x}{\cos x} \\ &= (\cos x) \log(\cos x) - \sin x \tan x \end{aligned}$$

Finally, we solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) [(\cos x) \log(\cos x) - \sin x \tan x] \\ &= (\cos x)^{\sin x} [(\cos x) \log(\cos x) - \sin x \tan x] \end{aligned}$$

Remark: negative numbers behave in a complicated manner when they are the base of an exponential expression. For example, the expression $(-1)^x$ is defined when x is the reciprocal of an odd number (like $x = \frac{1}{5}$ or $x = \frac{1}{7}$), but not when x is the reciprocal of an even number (like $x = \frac{1}{2}$). Since the domain of $f(x)$ was restricted to $(0, \frac{\pi}{2})$, $\cos x$ is always positive, and we avoid these complications.

S-25: Since $f(x)$ has the form of a function raised to a functional power, we will use logarithmic differentiation. We take the logarithm of the function, and make use of logarithm rules:

$$\log((\tan x)^x) = x \log(\tan x)$$

Now, we can differentiate:

$$\begin{aligned} \frac{\frac{d}{dx} \{(\tan x)^x\}}{(\tan x)^x} &= \log(\tan x) + x \cdot \frac{\sec^2 x}{\tan x} \\ &= \log(\tan x) + \frac{x}{\sin x \cos x} \end{aligned}$$

Finally, we solve for the derivative we want, $\frac{d}{dx}\{(\tan x)^x\}$:

$$\frac{d}{dx}\{(\tan x)^x\} = (\tan x)^x \left(\log(\tan x) + \frac{x}{\sin x \cos x} \right)$$

Remark: the restricted domain $(0, \pi/2)$ ensures that $\tan x$ is a positive number, so we avoid the problems that arise by raising a negative number to a variety of powers.

S-26: We use logarithmic differentiation.

$$\log f(x) = \log(x^2 + 1) \cdot (x^2 + 1)$$

We differentiate both sides to obtain:

$$\begin{aligned} \frac{f'(x)}{f(x)} &= \frac{d}{dx} \left\{ \log(x^2 + 1) \cdot (x^2 + 1) \right\} \\ &= \frac{2x}{x^2 + 1} (x^2 + 1) + 2x \log(x^2 + 1) \\ &= 2x(1 + \log(x^2 + 1)) \end{aligned}$$

Now, we solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) \cdot 2x(1 + \log(x^2 + 1)) \\ &= (x^2 + 1)^{x^2+1} \cdot 2x(1 + \log(x^2 + 1)) \end{aligned}$$

S-27: We use logarithmic differentiation: we modify our function to consider

$$\log f(x) = \log(x^2 + 1) \cdot \sin x$$

We differentiate using the product and chain rules:

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} \left\{ \log(x^2 + 1) \cdot \sin x \right\} = \cos x \cdot \log(x^2 + 1) + \frac{2x \sin x}{x^2 + 1}$$

Finally, we solve for $f'(x)$

$$\begin{aligned} f'(x) &= f(x) \cdot \left(\cos x \cdot \log(x^2 + 1) + \frac{2x \sin x}{x^2 + 1} \right) \\ &= (x^2 + 1)^{\sin(x)} \cdot \left(\cos x \cdot \log(x^2 + 1) + \frac{2x \sin x}{x^2 + 1} \right) \end{aligned}$$

S-28: We use logarithmic differentiation; so we modify our function to consider

$$\log f(x) = \log(x) \cdot \cos^3(x)$$

Differentiating, we find:

$$\frac{f'(x)}{f(x)} = \frac{d}{dx} \left\{ \log(x) \cdot \cos^3(x) \right\} = 3 \cos^2(x) \cdot (-\sin(x)) \cdot \log(x) + \frac{\cos^3(x)}{x}$$

Finally, we solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) \cdot \left(-3 \cos^2(x) \sin(x) \log(x) + \frac{\cos^3(x)}{x} \right) \\ &= x^{\cos^3(x)} \cdot \left(-3 \cos^2(x) \sin(x) \log(x) + \frac{\cos^3(x)}{x} \right) \end{aligned}$$

Remark: negative numbers behave in a complicated manner when they are the base of an exponential expression. For example, the expression $(-1)^x$ is defined when x is the reciprocal of an odd number (like $x = \frac{1}{5}$ or $x = \frac{1}{7}$), but not when x is the reciprocal of an even number (like $x = \frac{1}{2}$). Since the domain of $f(x)$ was restricted so that x is always positive, we avoid these complications.

S-29: We use logarithmic differentiation. So, we modify our function and consider

$$\log f(x) = (x^2 - 3) \cdot \log(3 + \sin(x)).$$

We differentiate:

$$\begin{aligned} \frac{f'(x)}{f(x)} &= \frac{d}{dx} \left\{ (x^2 - 3) \cdot \log(3 + \sin(x)) \right\} \\ &= 2x \log(3 + \sin(x)) + (x^2 - 3) \frac{\cos(x)}{3 + \sin(x)} \end{aligned}$$

Finally, we solve for $f'(x)$:

$$\begin{aligned} f'(x) &= f(x) \cdot \left[2x \log(3 + \sin(x)) + \frac{(x^2 - 3) \cos(x)}{3 + \sin(x)} \right] \\ &= (3 + \sin(x))^{x^2 - 3} \cdot \left[2x \log(3 + \sin(x)) + \frac{(x^2 - 3) \cos(x)}{3 + \sin(x)} \right] \end{aligned}$$

S-30: We will use logarithmic differentiation. First, we take the logarithm of our function, so we can use logarithm rules.

$$\log \left([f(x)]^{g(x)} \right) = g(x) \log(f(x))$$

Now, we differentiate. On the left side we use the chain rule, and on the right side we use product and chain rules.

$$\begin{aligned} \frac{d}{dx} \left\{ \log \left([f(x)]^{g(x)} \right) \right\} &= \frac{d}{dx} \{ g(x) \log(f(x)) \} \\ \frac{\frac{d}{dx} \{ [f(x)]^{g(x)} \}}{[f(x)]^{g(x)}} &= g'(x) \log(f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \end{aligned}$$

Finally, we solve for the derivative of our original function.

$$\frac{d}{dx}\{[f(x)]^{g(x)}\} = [f(x)]^{g(x)} \left(g'(x) \log(f(x)) + g(x) \cdot \frac{f'(x)}{f(x)} \right)$$

Remark: in this section, we have differentiated problems of this type several times—for example, Questions [24](#) through [29](#).

S-31: Let $g(x) := \log(f(x))$. Notice $g'(x) = \frac{f'(x)}{f(x)}$.

In order to show that the two curves have horizontal tangent lines at the same values of x , we will show two things: first, that if $f(x)$ has a horizontal tangent line at some value of x , then also $g(x)$ has a horizontal tangent line at that value of x . Second, we will show that if $g(x)$ has a horizontal tangent line at some value of x , then also $f(x)$ has a horizontal tangent line at that value of x .

Suppose $f(x)$ has a horizontal tangent line where $x = x_0$ for some point x_0 . This means $f'(x_0) = 0$. Then $g'(x_0) = \frac{f'(x_0)}{f(x_0)}$. Since $f(x_0) \neq 0$, $\frac{f'(x_0)}{f(x_0)} = \frac{0}{f(x_0)} = 0$, so $g(x)$ also has a horizontal tangent line when $x = x_0$. This shows that whenever f has a horizontal tangent line, g has one too.

Now suppose $g(x)$ has a horizontal tangent line where $x = x_0$ for some point x_0 . This means $g'(x_0) = 0$. Then $g'(x_0) = \frac{f'(x_0)}{f(x_0)} = 0$, so $f'(x_0)$ exists and is equal to zero. Therefore, $f(x)$ also has a horizontal tangent line when $x = x_0$. This shows that whenever g has a horizontal tangent line, f has one too.

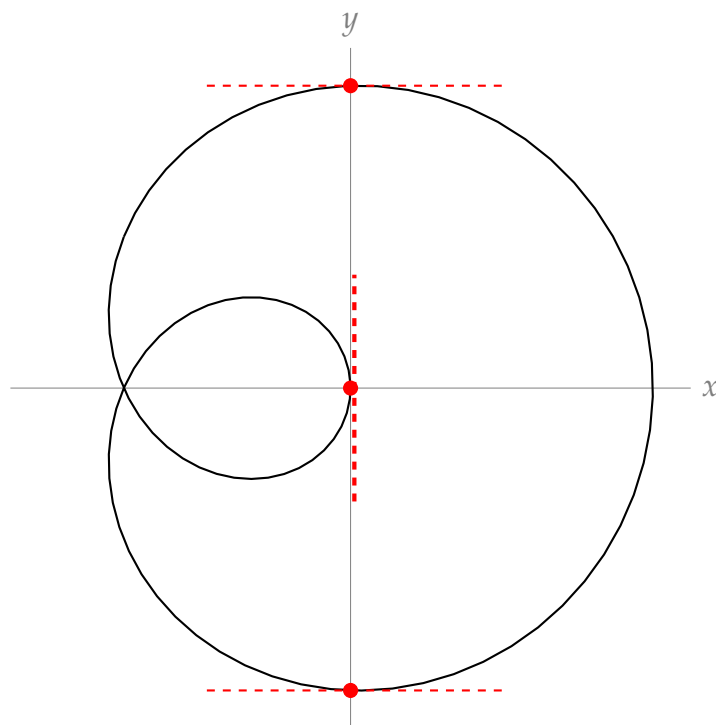
Remark: if we were not told that $f(x)$ gives only positive numbers, it would not necessarily be true that $f(x)$ and $\log(f(x))$ have horizontal tangent lines at the same values of x . If $f(x)$ had a horizontal tangent line at an x -value where $f(x)$ were negative, then $\log(f(x))$ would not exist there, let alone have a horizontal tangent line.

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S-1: We use the power rule (a) and the chain rule (b): the power rule tells us to “bring down the 2”, and the chain rule tells us to multiply by y' .

There is no need for the quotient rule here, as there are no quotients. Exponential functions have the form $(\text{constant})^{\text{function}}$, but our function has the form $(\text{function})^{\text{constant}}$, so we did not use (d).

S-2: At $(0, 4)$ and $(0, -4)$, the curve looks to be horizontal, if you zoom in: a tangent line here would have derivative zero. At the origin, the curve looks like its tangent line is vertical, so $\frac{dy}{dx}$ does not exist.



S-3: (a) No. A function must pass the vertical line test: one input cannot result in two (or more) outputs. Since one value of x sometimes corresponds to two values of y (for example, when $x = \pi/4$, y is $\pm 1/\sqrt{2}$), there is no function $f(x)$ so that $y = f(x)$ captures every point on the circle.

Remark: $y = \pm\sqrt{1-x^2}$ does capture every point on the unit circle. However, since one input x sometimes results in two outputs y , this expression is not a function.

(b) No, for the same reasons as (a). If $f'(x)$ is a function, then it can give at most one slope corresponding to one value of x . Since one value of x can correspond to two points on the circle with different slopes, $f'(x)$ cannot give the slope of every point on the circle. For example, fix any $0 < a < 1$. There are two points on the circle with x -coordinate equal to a . At the upper one, the slope is strictly negative. At the lower one, the slope is strictly positive.

(c) We differentiate:

$$2x + 2y \frac{dy}{dx} = 0$$

and solve for $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{y}$$

But there is a y in the right-hand side of this equation, and it's not clear how to get it out. Our answer in (b) tells us that, actually, we *can't* get it out, if we want the right-hand side to be a function of x . The derivative cannot be expressed as a function of x , because one value of x corresponds to multiple points on the circle.

Remark: since $y = \pm\sqrt{1-x^2}$, we could try writing

$$\frac{dy}{dx} = -\frac{x}{y} = \pm \frac{x}{\sqrt{1-x^2}}$$

but this is not a *function* of x . Again, in a function, one input leads to at most one output, but here one value of x will usually lead to two values of $\frac{dy}{dx}$.

S-4: Remember that y is a function of x . We begin with implicit differentiation.

$$\begin{aligned}xy + e^x + e^y &= 1 \\ y + x \frac{dy}{dx} + e^x + e^y \frac{dy}{dx} &= 0\end{aligned}$$

Now, we solve for $\frac{dy}{dx}$.

$$\begin{aligned}x \frac{dy}{dx} + e^y \frac{dy}{dx} &= -(e^x + y) \\ (x + e^y) \frac{dy}{dx} &= -(e^x + y) \\ \frac{dy}{dx} &= -\frac{e^x + y}{e^y + x}\end{aligned}$$

S-5: Differentiate both sides of the equation with respect to x :

$$e^y \frac{dy}{dx} = x \cdot 2y \frac{dy}{dx} + y^2 + 1$$

Now, get the derivative on one side and solve

$$\begin{aligned}e^y \frac{dy}{dx} - 2xy \frac{dy}{dx} &= y^2 + 1 \\ \frac{dy}{dx} (e^y - 2xy) &= y^2 + 1 \\ \frac{dy}{dx} &= \frac{y^2 + 1}{e^y - 2xy}\end{aligned}$$

S-6:

- First we find the x -coordinates where $y = 1$.

$$\begin{aligned}x^2 \tan\left(\frac{\pi}{4}\right) + 2x \log(1) &= 16 \\ x^2 \cdot 1 + 2x \cdot 0 &= 16 \\ x^2 &= 16\end{aligned}$$

So $x = \pm 4$.

- Now we use implicit differentiation to get y' in terms of x, y :

$$x^2 \tan(\pi y/4) + 2x \log(y) = 16$$

$$2x \tan(\pi y/4) + x^2 \frac{\pi}{4} \sec^2(\pi y/4) \cdot y' + 2 \log(y) + \frac{2x}{y} \cdot y' = 0.$$

- Now set $y = 1$ and use $\tan(\pi/4) = 1, \sec(\pi/4) = \sqrt{2}$ to get

$$2x \tan(\pi/4) + x^2 \frac{\pi}{4} \sec^2(\pi/4) y' + 2 \log(1) + 2x \cdot y' = 0$$

$$2x + \frac{\pi}{2} x^2 y' + 2x y' = 0$$

$$y' = -\frac{2x}{x^2 \pi/2 + 2x} = -\frac{4}{\pi x + 4}$$

- So at $(x, y) = (4, 1)$ we have $y' = -\frac{4}{4\pi + 4} = -\frac{1}{\pi + 1}$
- and at $(x, y) = (-4, 1)$ we have $y' = \frac{1}{\pi - 1}$

S-7: Differentiate the equation and solve:

$$3x^2 + 4y^3 \frac{dy}{dx} = -\sin(x^2 + y) \cdot \left(2x + \frac{dy}{dx}\right)$$

$$\frac{dy}{dx} = -\frac{2x \sin(x^2 + y) + 3x^2}{4y^3 + \sin(x^2 + y)}$$

S-8:

- First we find the x -coordinates where $y = 0$.

$$x^2 e^0 + 4x \cos(0) = 5$$

$$x^2 + 4x - 5 = 0$$

$$(x + 5)(x - 1) = 0$$

So $x = 1, -5$.

- Now we use implicit differentiation to get y' in terms of x, y :

$$x^2 e^y + 4x \cos(y) = 5 \quad \text{differentiate both sides}$$

$$x^2 \cdot e^y \cdot y' + 2x e^y + 4x(-\sin(y)) \cdot y' + 4 \cos(y) = 0$$

- Now set $y = 0$ to get

$$\begin{aligned}x^2 \cdot e^0 \cdot y' + 2xe^0 + 4x(-\sin(0)) \cdot y' + 4\cos(0) &= 0 \\x^2 y' + 2x + 4 &= 0 \\y' &= -\frac{4 + 2x}{x^2}.\end{aligned}$$

- So at $(x, y) = (1, 0)$ we have $y' = -6$,
- and at $(x, y) = (-5, 0)$ we have $y' = \frac{6}{25}$.

S-9: Differentiate the equation and solve:

$$\begin{aligned}2x + 2y \frac{dy}{dx} &= \cos(x + y) \cdot \left(1 + \frac{dy}{dx}\right) \\ \frac{dy}{dx} &= \frac{\cos(x + y) - 2x}{2y - \cos(x + y)}\end{aligned}$$

S-10:

- First we find the x -coordinates where $y = 0$.

$$\begin{aligned}x^2 \cos(0) + 2xe^0 &= 8 \\x^2 + 2x - 8 &= 0 \\(x + 4)(x - 2) &= 0\end{aligned}$$

So $x = 2, -4$.

- Now we use implicit differentiation to get y' in terms of x, y :

$$\begin{aligned}x^2 \cos(y) + 2xe^y &= 8 && \text{differentiate both sides} \\x^2 \cdot (-\sin y) \cdot y' + 2x \cos y + 2xe^y \cdot y' + 2e^y &= 0\end{aligned}$$

- Now set $y = 0$ to get

$$\begin{aligned}x^2 \cdot (-\sin 0) \cdot y' + 2x \cos 0 + 2xe^0 \cdot y' + 2e^0 &= 0 \\0 + 2x + 2xy' + 2 &= 0 \\y' &= -\frac{2 + 2x}{2x} = -\frac{1 + x}{x}\end{aligned}$$

- So at $(x, y) = (2, 0)$ we have $y' = -\frac{3}{2}$,
- and at $(x, y) = (-4, 0)$ we have $y' = -\frac{3}{4}$.

S-11: The question asks at which points on the ellipse $\frac{dy}{dx} = 1$. So, we begin by differentiating, implicitly:

$$2x + 6y \frac{dy}{dx} = 0$$

We could solve for $\frac{dy}{dx}$ at this point, but it's not necessary. We want to know when $\frac{dy}{dx}$ is equal to one:

$$\begin{aligned} 2x + 6y(1) &= 0 \\ x &= -3y \end{aligned}$$

That is, $\frac{dy}{dx} = 1$ at those points along the ellipse where $x = -3y$. We plug this into the equation of the ellipse to find the coordinates of these points.

$$\begin{aligned} (-3y)^2 + 3y^2 &= 1 \\ 12y^2 &= 1 \\ y &= \pm \frac{1}{\sqrt{12}} = \pm \frac{1}{2\sqrt{3}} \end{aligned}$$

So, the points along the ellipse where the tangent line is parallel to the line $y = x$ occur when $y = \frac{1}{2\sqrt{3}}$ and $x = -3y$, and when $y = \frac{-1}{2\sqrt{3}}$ and $x = -3y$. That is, the points $\left(\frac{-\sqrt{3}}{2}, \frac{1}{2\sqrt{3}}\right)$ and $\left(\frac{\sqrt{3}}{2}, \frac{-1}{2\sqrt{3}}\right)$.

S-12: First, we differentiate implicitly with respect to x .

$$\begin{aligned} \sqrt{xy} &= x^2y - 2 \\ \frac{1}{2\sqrt{xy}} \cdot \frac{d}{dx}\{xy\} &= (2x)y + x^2 \frac{dy}{dx} \\ \frac{y + x \frac{dy}{dx}}{2\sqrt{xy}} &= 2xy + x^2 \frac{dy}{dx} \end{aligned}$$

Now, we plug in $x = 1, y = 4$, and solve for $\frac{dy}{dx}$:

$$\begin{aligned} \frac{4 + \frac{dy}{dx}}{4} &= 8 + \frac{dy}{dx} \\ \frac{dy}{dx} &= -\frac{28}{3} \end{aligned}$$

S-13: Implicitly differentiating $x^2y(x)^2 + x \sin(y(x)) = 4$ with respect to x gives

$$2xy^2 + 2x^2yy' + \sin y + xy' \cos y = 0$$

Then we gather the terms containing y' on one side, so we can solve for y' :

$$\begin{aligned} 2x^2yy' + xy' \cos y &= -2xy^2 - \sin y \\ y'(2x^2y + x \cos y) &= -2xy^2 - \sin y \\ y' &= -\frac{2xy^2 + \sin y}{2x^2y + x \cos y} \end{aligned}$$

S-14:

- First we find the x -ordinates where $y = 0$.

$$\begin{aligned} x^2 + (1)e^0 &= 5 \\ x^2 + 1 &= 5 \\ x^2 &= 4 \end{aligned}$$

So $x = 2, -2$.

- Now we use implicit differentiation to get y' in terms of x, y :

$$2x + (y + 1)e^y \frac{dy}{dx} + e^y \frac{dy}{dx} = 0$$

- Now set $y = 0$ to get

$$\begin{aligned} 2x + (0 + 1)e^0 \frac{dy}{dx} + e^0 \frac{dy}{dx} &= 0 \\ 2x + \frac{dy}{dx} + \frac{dy}{dx} &= 0 \\ 2x &= -2 \frac{dy}{dx} \\ x &= -\frac{dy}{dx} \end{aligned}$$

- So at $(x, y) = (2, 0)$ we have $y' = -2$,
- and at $(x, y) = (-2, 0)$ we have $y' = 2$.

S-15: The slope of the tangent line is, of course, given by the derivative, so let's start by finding $\frac{dy}{dx}$ of both shapes.

For the circle, we differentiate implicitly

$$2x + 2y \frac{dy}{dx} = 0$$

and solve for $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{y}$$

For the ellipse, we also differentiate implicitly:

$$2x + 6y\frac{dy}{dx} = 0$$

and solve for $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{3y}$$

What we want is a value of x where both derivatives are equal. However, they might have different values of y , so let's let y_1 be the y -values associated with x on the circle, and let y_2 be the y -values associated with x on the ellipse. That is, $x^2 + y_1^2 = 1$ and $x^2 + 3y_2^2 = 1$. For the slopes at (x, y_1) on the circle and (x, y_2) on the ellipse to be equal, we need:

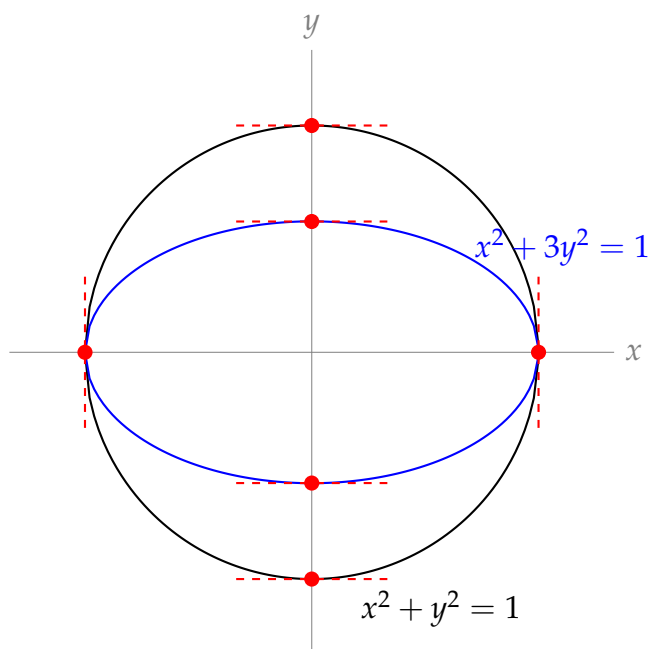
$$\begin{aligned} -\frac{x}{y_1} &= -\frac{x}{3y_2} \\ x \left(\frac{1}{y_1} - \frac{1}{3y_2} \right) &= 0 \end{aligned}$$

So $x = 0$ or $y_1 = 3y_2$. Let's think about which x -values will have a y -coordinate of the circle be three times as large as a y -coordinate of the ellipse. If $y_1 = 3y_2$, (x, y_1) is on the circle, and (x, y_2) is on the ellipse, then $x^2 + y_1^2 = x^2 + (3y_2)^2 = 1$ and $x^2 + 3y_2^2 = 1$. In this case:

$$\begin{aligned} x^2 + 9y_2^2 &= x^2 + 3y_2^2 \\ 9y_2^2 &= 3y_2^2 \\ y_2 &= 0 \\ x &= \pm 1 \end{aligned}$$

We need to be a tiny bit careful here: when $y = 0$, y' is not defined for either curve. For both curves, when $y = 0$, the tangent lines are vertical (and so have no real-valued slope!). Two vertical lines are indeed parallel.

So, for $x = 0$ and for $x = \pm 1$, the two curves have parallel tangent lines.



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S-1: (a) We can plug any number into the cosine function, and it will return a number in $[-1, 1]$. The domain of $\arcsin x$ is $[-1, 1]$, so any number we plug into cosine will give us a valid number to plug into arcsine. So, the domain of $f(x)$ is all real numbers.

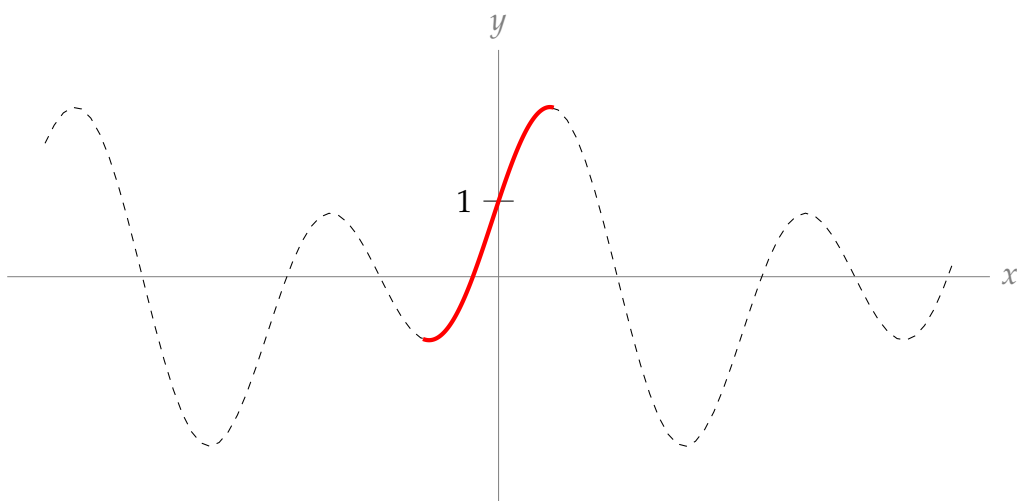
(b) We can plug any number into the cosine function, and it will return a number in $[-1, 1]$. The domain of $\operatorname{arccsc} x$ is $(-\infty, -1] \cup [1, \infty)$, so in order to have a valid number to plug into arccosecant, we need $\cos x = \pm 1$. That is, the domain of $g(x)$ is all values $x = n\pi$ for some integer n .

(c) The domain of arccosine is $[-1, 1]$. The domain of sine is all real numbers, so no matter what number arccosine spits out, we can safely plug it into sine. So, the domain of $h(x)$ is $[-1, 1]$.

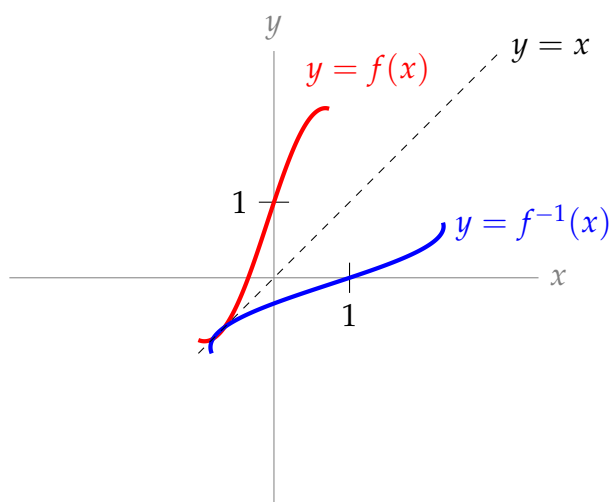
S-2: False: $\cos t = 1$ for infinitely many values of t ; arccosine gives only the single value $t = 0$ for which $\cos t = 1$ and $0 \leq t \leq \pi$. The particle does not start moving until $t = 10$, so $t = 0$ is not in the domain of the function describing its motion.

The particle will have height 1 at time $2\pi n$, for any integer $n \geq 2$.

S-3: First, we restrict the domain of f to force it to be one-to-one. There are many intervals we could choose over which f is one-to-one, but the question asks us to contain $x = 0$ and be as large as possible; this leaves us with the following restricted function:



The inverse of a function swaps the role of the input and output; so if the graph of $y = f(x)$ contains the point (a, b) , then the graph of $Y = f^{-1}(X)$ contains the point (b, a) . That is, the graph of $Y = f^{-1}(X)$ is the graph of $y = f(x)$ with the x -coordinates and y -coordinates swapped. (So, since $y = f(x)$ crosses the y -axis at $y = 1$, then $Y = f^{-1}(X)$ crosses the X -axis at $X = 1$.) This swapping is equivalent to reflecting the curve $y = f(x)$ over the line $y = x$.



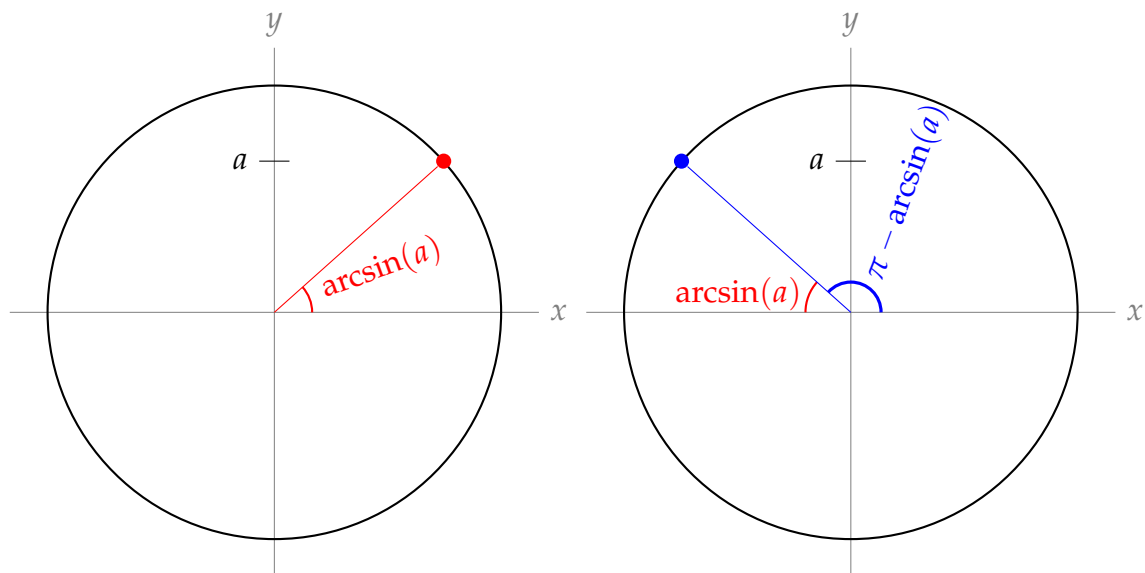
Remark: while you're getting accustomed to inverse functions, it is sometimes clearer to consider $y = f(x)$ and $Y = f^{-1}(X)$: using slightly different notations for x (the input of f , hence the output of f^{-1}) and X (the input of f^{-1} , which comes from the output of f). However, the convention is to use x for the inputs of both functions, and y as the outputs of both functions, as is written on the graph above.

S-4: The tangent line is horizontal when $0 = y' = a - \sin x$. That is, when $a = \sin x$.

- If $|a| > 1$, then there is no value of x for which $a = \sin x$, so the curve has no horizontal tangent lines.
- If $|a| = 1$, then there are infinitely many solutions to $a = \sin x$, but only one solution in the interval $[-\pi, \pi]$: $x = \arcsin(a) = \arcsin(\pm 1) = \pm \frac{\pi}{2}$. Then the values of x for

which $a = \sin x$ are $x = 2\pi n + a\frac{\pi}{2}$ for any integer n .

- If $|a| < 1$, then there are infinitely many solutions to $a = \sin x$. The solution in the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ is given by $x = \arcsin(a)$. The other solution in the interval $(-\pi, \pi)$ is given by $x = \pi - \arcsin(a)$, as shown in the unit circles below.



So, the values of x for which $x = \sin a$ are $x = 2\pi n + \arcsin(a)$ and $x = 2\pi n + \pi - \arcsin(a)$ for any integer n .

Remark: when $a = 1$, then

$$2\pi n + \arcsin(a) = 2\pi n + \frac{\pi}{2} = 2\pi n + \pi - \left(\frac{\pi}{2}\right) = 2\pi n + \pi - \arcsin(a).$$

Similarly, when $a = -1$,

$$2\pi n + \arcsin(a) = 2\pi n - \frac{\pi}{2} = 2\pi(n-1) + \pi - \left(-\frac{\pi}{2}\right) = 2\pi(n-1) + \pi - \arcsin(a).$$

So, if we try to use the descriptions in the third bullet point to describe points where the tangent line is horizontal when $|a| = 1$, we get the correct points but each point is listed twice. This is why we separated the case $|a| = 1$ from the case $|a| < 1$.

S-5: The function $\arcsin x$ is only defined for $|x| \leq 1$, and the function $\operatorname{arccsc} x$ is only defined for $|x| \geq 1$, so $f(x)$ has domain $|x| = 1$. That is, $x = \pm 1$.

In order for $f(x)$ to be differentiable at a point, it must exist in an open interval around that point. (See Definition 2.2.1.) Since our function does not exist over any open interval, $f(x)$ is not differentiable anywhere.

So, actually, $f(x)$ is a pretty boring function, which we can entirely describe as:
 $f(-1) = -\pi$ and $f(1) = \pi$.

S-6: Using the chain rule,

$$\begin{aligned}\frac{d}{dx} \left\{ \arcsin \left(\frac{x}{3} \right) \right\} &= \frac{1}{\sqrt{1 - \left(\frac{x}{3} \right)^2}} \cdot \frac{1}{3} \\ &= \frac{1}{3\sqrt{1 - \frac{x^2}{9}}} \\ &= \frac{1}{\sqrt{9 - x^2}}\end{aligned}$$

Since the domain of arcsine is $[-1, 1]$, and we are plugging in $\frac{x}{3}$ to arcsine, the values of x that we can plug in are those that satisfy $-1 \leq \frac{x}{3} \leq 1$, or $-3 \leq x \leq 3$. So the domain of f is $[-3, 3]$.

S-7: Using the quotient rule,

$$\frac{d}{dt} \left\{ \frac{\arccos t}{t^2 - 1} \right\} = \frac{(t^2 - 1) \left(\frac{-1}{\sqrt{1 - t^2}} \right) - (\arccos t)(2t)}{(t^2 - 1)^2}$$

The domain of arccosine is $[-1, 1]$, and since $t^2 - 1$ is in the denominator, the domain of f requires $t^2 - 1 \neq 0$, that is, $t \neq \pm 1$. So the domain of $f(t)$ is $(-1, 1)$.

S-8: The domain of arcsec x is $|x| \geq 1$: that is, we can plug into arcsecant only values with absolute value greater than or equal to one. Since $-x^2 - 2 \leq -2$, every real value of x gives us an acceptable value to plug into arcsecant. So, the domain of $f(x)$ is all real numbers.

To differentiate, we use the chain rule. Remember $\frac{d}{dx} \{\operatorname{arcsec} x\} = \frac{1}{|x|\sqrt{x^2 - 1}}$.

$$\begin{aligned}\frac{d}{dx} \left\{ \operatorname{arcsec}(-x^2 - 2) \right\} &= \frac{1}{|-x^2 - 2|\sqrt{(-x^2 - 2)^2 - 1}} \cdot (-2x) \\ &= \frac{-2x}{(x^2 + 2)\sqrt{x^4 + 4x^2 + 3}}.\end{aligned}$$

S-9: We use the chain rule, remembering that a is a constant.

$$\begin{aligned}\frac{d}{dx} \left\{ \frac{1}{a} \arctan \left(\frac{x}{a} \right) \right\} &= \frac{1}{a} \cdot \frac{1}{1 + \left(\frac{x}{a} \right)^2} \cdot \frac{1}{a} \\ &= \frac{1}{a^2 + x^2}\end{aligned}$$

The domain of arctangent is all real numbers, so the domain of $f(x)$ is also all real numbers.

S-10: We differentiate using the **product** and **chain** rules.

$$\begin{aligned}\frac{d}{dx} \left\{ x \arcsin x + \sqrt{1-x^2} \right\} &= \arcsin x + \frac{x}{\sqrt{1-x^2}} + \frac{-2x}{2\sqrt{1-x^2}} \\ &= \arcsin x\end{aligned}$$

The domain of $\arcsin x$ is $[-1, 1]$, and the domain of $\sqrt{1-x^2}$ is all values of x so that $1-x^2 \geq 0$, so x in $[-1, 1]$. Therefore, the domain of $f(x)$ is $[-1, 1]$.

S-11: We differentiate using the chain rule:

$$\frac{d}{dx} \{\arctan(x^2)\} = \frac{2x}{1+x^4}$$

This is zero exactly when $x = 0$.

S-12: Using formulas you should memorize from this section,

$$\frac{d}{dx} \{\arcsin x + \arccos x\} = \frac{1}{\sqrt{1-x^2}} + \frac{-1}{\sqrt{1-x^2}} = 0$$

Remark: the only functions with derivative equal to zero everywhere are constant functions, so $\arcsin x + \arccos x$ should be a constant. Since $\sin \theta = \cos(\frac{\pi}{2} - \theta)$, we can set

$$\sin \theta = x \qquad \cos\left(\frac{\pi}{2} - \theta\right) = x$$

where x and θ are the same in both expressions, and $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then

$$\arcsin x = \theta \qquad \arccos x = \frac{\pi}{2} - \theta$$

We note here that arcsine is the inverse of the sine function *restricted to* $[-\frac{\pi}{2}, \frac{\pi}{2}]$. So, since we restricted θ to this domain, $\sin \theta = x$ really does imply $\arcsin x = \theta$. (For an example of why this matters, note $\sin(2\pi) = 0$, but $\arcsin(0) = 0 \neq 2\pi$.) Similarly, arccosine is the inverse of the cosine function restricted to $[0, \pi]$. Since $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, then $0 \leq (\frac{\pi}{2} - \theta) \leq \pi$, so $\cos(\frac{\pi}{2} - \theta) = x$ really does imply $\arccos x = \frac{\pi}{2} - \theta$.

So,

$$\arcsin x + \arccos x = \theta + \frac{\pi}{2} - \theta = \frac{\pi}{2}$$

which means the derivative we were calculating was actually just $\frac{d}{dx} \left\{ \frac{\pi}{2} \right\} = 0$.

S-13: Using the chain rule,

$$y' = \frac{-\frac{1}{x^2}}{\sqrt{1-\left(\frac{1}{x}\right)^2}} = \frac{-1}{x^2 \sqrt{1-\frac{1}{x^2}}}.$$

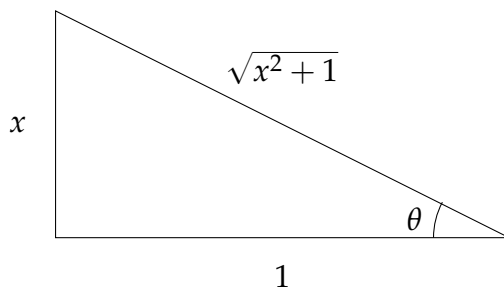
S-14: Using the chain rule,

$$y' = \frac{-\frac{1}{x^2}}{1 + \left(\frac{1}{x}\right)^2} = \frac{-1}{x^2 + 1}.$$

S-15: Using the product rule:

$$\begin{aligned}\frac{d}{dx} \left\{ (1 + x^2) \arctan x \right\} &= 2x \arctan x + (1 + x^2) \frac{1}{1 + x^2} \\ &= 2x \arctan x + 1\end{aligned}$$

S-16: Let $\theta = \arctan x$. Then θ is the angle of a right triangle that gives $\tan \theta = x$. In particular, the ratio of the opposite side to the adjacent side is x . So, we have a triangle that looks like this:



where the length of the hypotenuse came from the Pythagorean Theorem. Now,

$$\sin(\arctan x) = \sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{x}{\sqrt{x^2 + 1}}$$

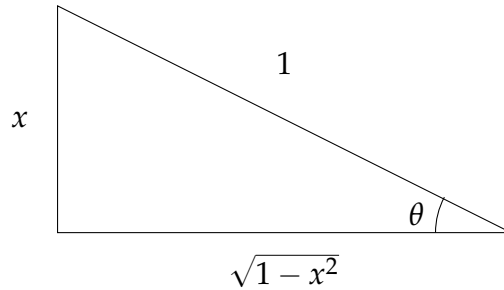
From here, we differentiate using the quotient rule:

$$\begin{aligned}\frac{d}{dx} \left\{ \frac{x}{\sqrt{x^2 + 1}} \right\} &= \frac{\sqrt{x^2 + 1} - x \frac{2x}{2\sqrt{x^2 + 1}}}{x^2 + 1} \\ &= \left(\frac{\sqrt{x^2 + 1} - \frac{x^2}{\sqrt{x^2 + 1}}}{x^2 + 1} \right) \cdot \frac{\sqrt{x^2 + 1}}{\sqrt{x^2 + 1}} \\ &= \frac{(x^2 + 1) - x^2}{(x^2 + 1)^{3/2}} \\ &= \frac{1}{(x^2 + 1)^{3/2}} = (x^2 + 1)^{-3/2}\end{aligned}$$

Remark: another strategy is to differentiate first, using the chain rule, then draw a triangle to simplify the resulting expression $\frac{d}{dx} \{ \sin(\arctan x) \} = \frac{\cos(\arctan x)}{1 + x^2}$.

S-17:

Let $\theta = \arcsin x$. Then θ is the angle of a right triangle that gives $\sin \theta = x$. In particular, the ratio of the opposite side to the hypotenuse is x . So, we have a triangle that looks like this:



where the length of the adjacent side came from the Pythagorean Theorem. Now,

$$\cot(\arcsin x) = \cot \theta = \frac{\text{adj}}{\text{opp}} = \frac{\sqrt{1-x^2}}{x}$$

From here, we differentiate using the quotient rule:

$$\begin{aligned} \frac{d}{dx} \left\{ \frac{\sqrt{1-x^2}}{x} \right\} &= \frac{x \frac{-2x}{2\sqrt{1-x^2}} - \sqrt{1-x^2}}{x^2} \\ &= \frac{-x^2 - (1-x^2)}{x^2 \sqrt{1-x^2}} \\ &= \frac{-1}{x^2 \sqrt{1-x^2}} \end{aligned}$$

Remark: another strategy is to differentiate first, using the chain rule, then draw a triangle to simplify the resulting expression $\frac{d}{dx} \{\cot(\arcsin x)\} = \frac{-\csc^2(\arcsin x)}{\sqrt{1-x^2}}$.

S-18: The line $y = 2x + 9$ has slope 2, so we must find all values of x between -1 and 1 ($\arcsin x$ is only defined for these values of x) for which $\frac{d}{dx} \{\arcsin x\} = 2$. Evaluating the

derivative:

$$\begin{aligned}y &= \arcsin x \\2 &= y' = \frac{1}{\sqrt{1-x^2}} \\4 &= \frac{1}{1-x^2} \\\frac{1}{4} &= 1-x^2 \\x^2 &= \frac{3}{4} \\x &= \pm \frac{\sqrt{3}}{2} \\(x, y) &= \pm \left(\frac{\sqrt{3}}{2}, \frac{\pi}{3}\right)\end{aligned}$$

S-19: We differentiate using the chain rule:

$$\begin{aligned}\frac{d}{dx}\{\arctan(\csc x)\} &= \frac{1}{1+\csc^2 x} \cdot \frac{d}{dx}\{\csc x\} \\&= \frac{-\csc x \cot x}{1+\csc^2 x} \\&= \frac{-\frac{1}{\sin x} \cdot \frac{\cos x}{\sin x}}{1+\left(\frac{1}{\sin x}\right)^2} \\&= \frac{-\cos x}{\sin^2 x + 1}\end{aligned}$$

So if $f'(x) = 0$, then $\cos x = 0$, and this happens when $x = \frac{(2n+1)\pi}{2}$ for any integer n . We should check that these points are in the domain of f . Arctangent is defined for all real numbers, so we only need to check the domain of cosecant; when $x = \frac{(2n+1)\pi}{2}$, then $\sin x = \pm 1 \neq 0$, so $\csc x = \frac{1}{\sin x}$ exists.

S-20: Since $g(y) = f^{-1}(y)$,

$$f(g(y)) = f(f^{-1}(y)) = y$$

Now, we can differentiate with respect to y using the chain rule.

$$\begin{aligned}\frac{d}{dy}\{f(g(y))\} &= \frac{d}{dy}\{y\} \\f'(g(y)) \cdot g'(y) &= 1 \\g'(y) &= \frac{1}{f'(g(y))} = \frac{1}{1 - \sin g(y)}\end{aligned}$$

S-21: Write $g(y) = f^{-1}(y)$. Then $g(f(x)) = x$, so differentiating both sides (using the chain rule), we see

$$g'(f(x)) \cdot f'(x) = 1$$

What we want is $g'(\pi - 1)$, so we need to figure out which value of x gives $f(x) = \pi - 1$. A little trial and error leads us to $x = \frac{\pi}{2}$.

$$g'(\pi - 1) \cdot f'\left(\frac{\pi}{2}\right) = 1$$

Since $f'(x) = 2 - \cos(x)$, $f'\left(\frac{\pi}{2}\right) = 2 - 0 = 2$:

$$\begin{aligned} g'(\pi - 1) \cdot 2 &= 1 \\ g'(\pi - 1) &= \frac{1}{2} \end{aligned}$$

S-22: Write $g(y) = f^{-1}(y)$. Then $g(f(x)) = x$, so differentiating both sides (using the chain rule), we see

$$g'(f(x))f'(x) = 1$$

What we want is $g'(e + 1)$, so we need to figure out which value of x gives $f(x) = e + 1$. A little trial and error leads us to $x = 1$.

$$\begin{aligned} g'(f(1))f'(1) &= 1 \\ g'(e + 1) \cdot f'(1) &= 1 \\ g'(e + 1) &= \frac{1}{f'(1)} \end{aligned}$$

It remains only to note that $f'(x) = e^x + 1$, so $f'(1) = e + 1$

$$g'(e + 1) = \frac{1}{e + 1}$$

S-23: We use logarithmic differentiation, our standard method of differentiating an expression of the form (function)^{function}.

$$\begin{aligned} f(x) &= [\sin x + 2]^{\operatorname{arcsec} x} \\ \log(f(x)) &= \operatorname{arcsec} x \cdot \log[\sin x + 2] \\ \frac{f'(x)}{f(x)} &= \frac{1}{|x|\sqrt{x^2 - 1}} \log[\sin x + 2] + \operatorname{arcsec} x \cdot \frac{\cos x}{\sin x + 2} \\ f'(x) &= [\sin x + 2]^{\operatorname{arcsec} x} \left(\frac{\log[\sin x + 2]}{|x|\sqrt{x^2 - 1}} + \frac{\operatorname{arcsec} x \cdot \cos x}{\sin x + 2} \right) \end{aligned}$$

The domain of $\operatorname{arcsec} x$ is $|x| \geq 1$. For any x , $\sin x + 2$ is positive, and a positive number can be raised to any power. (Recall negative numbers cannot be raised to any power—for example, $(-1)^{1/2} = \sqrt{-1}$ is not a real number.) So, the domain of $f(x)$ is $|x| \geq 1$.

S-24: The function $\frac{1}{\sqrt{x^2 - 1}}$ exists only for those values of x with $x^2 - 1 > 0$: that is, the domain of $\frac{1}{\sqrt{x^2 - 1}}$ is $|x| > 1$. However, the domain of arcsine is $|x| \leq 1$. So, there is not one single value of x where $\arcsin x$ and $\frac{1}{\sqrt{x^2 - 1}}$ are both defined.

If the derivative of $\arcsin(x)$ were given by $\frac{1}{\sqrt{x^2 - 1}}$, then the derivative of $\arcsin(x)$ would not exist anywhere, so we would probably just write “derivative does not exist,” instead of making up a function with a mismatched domain. Also, the function $f(x) = \arcsin(x)$ is a smooth curve—its derivative exists at every point strictly inside its domain. (Remember not all curves are like this: for instance, $g(x) = |x|$ does not have a derivative at $x = 0$, but $x = 0$ is strictly inside its domain.) So, it’s a pretty good bet that the derivative of arcsine is *not* $\frac{1}{\sqrt{x^2 - 1}}$.

S-25: This limit represents the derivative computed at $x = 1$ of the function $f(x) = \arctan x$. To see this, simply use the definition of the derivative at $a = 1$:

$$\begin{aligned} \left. \frac{d}{dx} \{f(x)\} \right|_a &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} \\ \left. \frac{d}{dx} \{\arctan x\} \right|_1 &= \lim_{x \rightarrow 1} \frac{\arctan x - \arctan 1}{x - 1} \\ &= \lim_{x \rightarrow 1} \frac{\arctan x - \frac{\pi}{4}}{x - 1} \\ &= \lim_{x \rightarrow 1} \left((x - 1)^{-1} \left(\arctan x - \frac{\pi}{4} \right) \right). \end{aligned}$$

Since the derivative of $f(x)$ is $\frac{1}{1 + x^2}$, its value at $x = 1$ is exactly $\frac{1}{2}$.

S-26: First, let’s interpret the given information: when the input of our function is $2x + 1$ for some x , then its output is $\frac{5x - 9}{3x + 7}$, for that same x . We’re asked to evaluate $f^{-1}(7)$, which is the number y with the property that $f(y) = 7$. If the output of our function is 7, that means

$$7 = \frac{5x - 9}{3x + 7}$$

and so

$$\begin{aligned} 7(3x + 7) &= 5x - 9 \\ x &= -\frac{29}{8} \end{aligned}$$

So, when $x = -\frac{29}{8}$, our equation $f(2x + 1) = \frac{5x - 9}{3x + 7}$ becomes:

$$f\left(2 \cdot \frac{-29}{8} + 1\right) = \frac{5 \cdot \frac{-29}{8} - 9}{3 \cdot \frac{-29}{8} + 7}$$

Or, equivalently:

$$f\left(-\frac{25}{4}\right) = 7$$

Therefore, $f^{-1}(7) = -\frac{25}{4}$.

S-27: If $f^{-1}(y) = 0$, that means $f(0) = y$. So, we want to find out what we plug into f^{-1} to get 0. Since we only know f^{-1} in terms of a variable x , let's figure out what x gives us an output of 0:

$$\begin{aligned}\frac{2x + 3}{x + 1} &= 0 \\ 2x + 3 &= 0 \\ x &= -\frac{3}{2}\end{aligned}$$

Now, the equation $f^{-1}(4x - 1) = \frac{2x + 3}{x + 1}$ with $x = -\frac{3}{2}$ tells us:

$$f^{-1}\left(4 \cdot \frac{-3}{2} - 1\right) = \frac{2 \cdot \frac{-3}{2} + 3}{\frac{-3}{2} + 1}$$

Or, equivalently:

$$f^{-1}(-7) = 0$$

Therefore, $f(0) = -7$.

S-28:

- Solution 1: We begin by differentiating implicitly. Following the usual convention,

we use y' to mean $y'(x)$.

$$\begin{aligned}
 \arcsin(x + 2y) &= x^2 + y^2 && \text{Using the chain rule:} \\
 \frac{1 + 2y'}{\sqrt{1 - (x + 2y)^2}} &= 2x + 2yy' \\
 \frac{1}{\sqrt{1 - (x + 2y)^2}} + \frac{2y'}{\sqrt{1 - (x + 2y)^2}} &= 2x + 2yy' \\
 \frac{2y'}{\sqrt{1 - (x + 2y)^2}} - 2yy' &= 2x - \frac{1}{\sqrt{1 - (x + 2y)^2}} \\
 y' \left(\frac{2}{\sqrt{1 - (x + 2y)^2}} - 2y \right) &= 2x - \frac{1}{\sqrt{1 - (x + 2y)^2}} \\
 y' &= \frac{2x - \frac{1}{\sqrt{1 - (x + 2y)^2}}}{\frac{2}{\sqrt{1 - (x + 2y)^2}} - 2y} \left(\frac{\sqrt{1 - (x + 2y)^2}}{\sqrt{1 - (x + 2y)^2}} \right) \\
 y' &= \frac{2x\sqrt{1 - (x + 2y)^2} - 1}{2 - 2y\sqrt{1 - (x + 2y)^2}}
 \end{aligned}$$

- Solution 2: We begin by taking the sine of both sides of the equation.

$$\begin{aligned}
 \arcsin(x + 2y) &= x^2 + y^2 \\
 x + 2y &= \sin(x^2 + y^2)
 \end{aligned}$$

Now, we differentiate implicitly.

$$\begin{aligned}
 1 + 2y' &= \cos(x^2 + y^2) \cdot (2x + 2yy') \\
 1 + 2y' &= 2x \cos(x^2 + y^2) + 2yy' \cos(x^2 + y^2) \\
 2y' - 2yy' \cos(x^2 + y^2) &= 2x \cos(x^2 + y^2) - 1 \\
 y' (2 - 2y \cos(x^2 + y^2)) &= 2x \cos(x^2 + y^2) - 1 \\
 y' &= \frac{2x \cos(x^2 + y^2) - 1}{2 - 2y \cos(x^2 + y^2)}
 \end{aligned}$$

- We used two different methods, and got two answers that look pretty different. However, the answers ought to be equivalent. To see this, we remember that for all values of x and y that we care about (those pairs (x, y) in the domain of our curve), the equality

$$\arcsin(x + 2y) = x^2 + y^2$$

holds. Drawing a triangle:
